Lecture 7: Bayesian Games, Adverse Selection and Auctions

Detailed solutions to the 24 items of `Listas/pratica-lecture-07.md`
Economics of Organizations and Incentives (MPE Insper)

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Each item follows the list's order and numbering. Each statement is printed verbatim from the list right before its solution. Every number and equilibrium claim is asserted in `check_E1.py` (items 1–8), `check_E2.py` (9–18) and `check_E3.py` (19–24), run from this folder. Notation: a type $t_i \in T_i$ is drawn by Nature from a common prior; a strategy is a function $s_i : T_i \to A_i$; a profile s^* is a Bayesian Nash equilibrium (BNE) if every type t_i of every player maximizes his expected payoff given his posterior over t_{-i} and the others' strategies (Gibbons §3.1.B, pp. 149–150; Tadelis §12.1, pp. 246–252). The unit is estimated: there are no Lecture 7 slides yet, so all references are to the books.

Contents

Lecture 7 (estimated) — Bayesian games, adverse selection and auctions 2

Item 1: Gibbons Problem 3.1 (p. 169) 2

Item 2: Muñoz-Garcia Ch. 7, Exercise 1 (p. 318–319) 3

Item 3: Gibbons Problem 3.4 (p. 169–170) 5

Item 4: Tadelis Exercise 12.1 (p. 266) 7

Item 5: Tadelis Exercise 13.1 (p. 285) 10

Item 6: Muñoz-Garcia Ch. 8, Exercise 4 (p. 362) 11

Item 7: Muñoz-Garcia Ch. 8, Exercise 1 (p. 350) 13

Item 8: Tadelis Exercise 13.2 (p. 285) 14

Item 9: Tadelis Exercise 12.2 (p. 266) 15

Item 10: Gibbons Problem 3.3 (p. 169) 16

Item 11: Tadelis Exercise 12.3 (p. 266–267) 18

Item 12: Tadelis Exercise 12.9 (p. 269) 20

Item 13: Tadelis Exercise 12.5 (p. 267–268) **22**

Item 14: Tadelis Exercise 12.6 (p. 268) **24**

Item 15: Gibbons Problem 3.6 (p. 170) **26**

Item 16: Tadelis Exercise 13.9 (p. 287) **27**

Item 17: Tadelis Exercise 13.3 (p. 285–286) **28**

Item 18: Tadelis Exercise 12.8 (p. 268–269) **30**

Item 19: MWG Exercise 13.B.1 (p. 473) **33**

Item 20: MWG Exercise 13.B.2 (p. 473) **35**

Item 21: Tadelis Exercise 12.7 (p. 268) **37**

Item 22: Gibbons Problem 3.8 (p. 171–172) **39**

Item 23: Tadelis Exercise 13.5 (p. 286) **42**

Item 24: Tadelis Exercise 13.10 (p. 287) **44**

Lecture 7 (estimated) — Bayesian games, adverse selection and auctions

Item 1: Gibbons Problem 3.1

Foundations · difficulty 1/5

Statement (verbatim from the list, item 1)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §3.5 Problems, Section 3.1, Problem 3.1, p. 169 [PDF p. 175]. Reproduced verbatim. **Answer in the source:** no.

> **3.1.** What is a static Bayesian game? What is a (pure) strategy in such a game? What is a (pure-strategy) Bayesian Nash equilibrium in such a game?

(a) What is asked. Define a static Bayesian game, a pure strategy in it, and a pure-strategy Bayesian Nash equilibrium.

(b) Method. Gibbons §3.1.B (pp. 149–151); Tadelis §12.1 (pp. 246–252). Information is incomplete because a player's payoff depends on a parameter (his *type*) that only he observes; Harsanyi's device makes Nature draw the types first, which turns the game into one of imperfect information.

(c) Solution. **Static Bayesian game.** $G = \{A_1, \ldots, A_n; T_1, \ldots, T_n; p_1, \ldots, p_n; u_1, \ldots, u_n\}$: action spaces A_i; type spaces T_i; beliefs $p_i(t_{-i} \mid t_i)$ (from a common prior, so that p_i is the conditional of one joint distribution on T); payoffs $u_i(a_1, \ldots, a_n; t_i)$. Timing: (1) Nature draws $t = (t_1, \ldots, t_n)$ and tells player i only t_i; (2) players choose actions $a_i \in A_i$ simultaneously; (3) payoffs u_i are realized.

Pure strategy. A function $s_i : T_i \to A_i$: it says which action each type of i takes. The strategy space has $|A_i|^{|T_i|}$ elements, and it is what a player with $|T_i|$ types is "really" choosing, even though only one type is realized.

Pure-strategy BNE. A profile $s^* = (s_1^*, \ldots, s_n^*)$ such that for every player i and every type $t_i \in T_i$, the action $s_i^*(t_i)$ solves

$$\max_{a_i \in A_i} \sum_{t_{-i} \in T_{-i}} u_i\big(s_1^*(t_1), \ldots, a_i, \ldots, s_n^*(t_n); t_i\big)\, p_i(t_{-i} \mid t_i).$$

Each type best-responds to the others' *strategies* (functions of their types) averaged over his beliefs. Because every type has positive probability, this interim condition is equivalent to Nash equilibrium of the Bayesian normal form, where player i picks a whole function s_i to maximize ex-ante expected payoff (Tadelis p. 251; Gibbons p. 150).

(e) Verification. `check_E1.py` builds the Bayesian normal form of the game in item 2 and checks that its Nash equilibria are exactly the profiles in which every type passes the interim best-reply test, and that the strategy count is $|A_i|^{|T_i|}$ ($2^2 = 4$ there). Gibbons gives no answer.

Answer. A static Bayesian game is $\{A_i; T_i; p_i; u_i\}$ with Nature drawing types first; a pure strategy is a map $s_i : T_i \to A_i$; a BNE is a profile in which every type of every player maximizes expected payoff given his belief $p_i(\cdot \mid t_i)$ and the others' strategies.

Item 2: Muñoz-Garcia Ch. 7, Exercise 1

Foundations · difficulty 1/5

Statement (verbatim from the list, item 2)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 7, Exercise 1 "A Simple Bayesian Game", pp. 318–319 [PDF pp. 329–330]. Reproduced verbatim (statement only; the book's worked answer follows it). **Answer in the source:** yes, the book's worked answer follows the exercise.

> Consider the payoff matrix in Fig. 7.1, where player 1 (in rows) chooses between A and B, and player 2 (in columns) selects C or D. Player 1 observes the realization of random variable x, which takes value $x = 15$ with probability 1/2 or $x = 4$ with probability 1/2. Player 2, however, doesn't observe the realization of x, but knows its probability distribution. Therefore, player 1 is perfectly informed about the payoff matrix they face, while player 2 does not know the payoff that player 1 earns in strategy profile (A, C). While this payoff only affects player 1, player 2's behavior is affected by this uncertainty since he could anticipate that player 1 best responds to column C with A when $x = 15$, but player 1 would best respond with B when $x = 4$.
>
> | Player 1 Player 2 | C | D |
> |---|---|---|
> | A | x, 18 | 6, 7 |
> | B | 8, 0 | 8, 11 |
>
> *Fig. 7.1* Simultaneous-move game under asymmetric information about x
>
> (a) *Complete information.* As a benchmark, let us assume in this part of the exercise that both players are informed about the realization of x. Which are the pure strategy Nash equilibria of the game when $x = 15$? And what are the pure strategy Nash equilibria when $x = 4$?
>
> (b) *Incomplete information.* Let us now return to the game where only player 1 privately observes the realization of x. Find the Bayesian Nash Equilibrium of this incomplete information game by following the next three steps:
>
> 1. Identify each player's strategy space in this incomplete information setting,
> 2. Construct the Bayesian normal form representation, and
> 3. Underline each player's best response payoffs to find the BNE of the game.

(a) What is asked. (a) The pure Nash equilibria of the matrix in Fig. 7.1 when $x = 15$ and when $x = 4$ is common knowledge. (b) The BNE when only player 1 observes x (prior $\frac{1}{2}, \frac{1}{2}$): strategy spaces, Bayesian normal form, best-response underlining.

(b) Method. (a) Underline best replies in each 2×2 matrix. (b) Player 1 has two types ($x = 15$, $x = 4$) and two actions, so four strategies $s_1 = (a^{15}, a^4)$; player 2 has one type and two actions. Expected payoffs: average the two realizations of x with weight $\frac{1}{2}$ (Tadelis §12.1; Gibbons §3.1.B).

(c) Solution. (a) Complete information. Player 2's payoffs do not depend on x: $BR_2(A) = C$ ($18 > 7$) and $BR_2(B) = D$ ($11 > 0$). For player 1, $BR_1(D) = B$ ($8 > 6$) for every x; against C he compares x with 8:

- $x = 15$: $BR_1(C) = A$ ($15 > 8$). Nash equilibria (A, C) and (B, D).
- $x = 4$: $BR_1(C) = B$ ($8 > 4$). The only Nash equilibrium is (B, D).

(b) Incomplete information. $S_1 = \{A^{15}A^4, A^{15}B^4, B^{15}A^4, B^{15}B^4\}$ (superscript = value of x observed), $S_2 = \{C, D\}$. The Bayesian normal form has $EU_1 = \frac{1}{2}u_1(\cdot;15) + \frac{1}{2}u_1(\cdot;4)$ and EU_2 the average over the realized x of player 2's payoff; e.g. $EU_1(A^{15}B^4, C) = \frac{1}{2}\cdot 15 + \frac{1}{2}\cdot 8 = 11.5$ and $EU_2(A^{15}B^4, C) = \frac{1}{2}\cdot 18 + \frac{1}{2}\cdot 0 = 9$. Best replies underlined:

	C	D
$A^{15}A^4$	9.5, <u>18</u>	6, 7
$A^{15}B^4$	<u>11.5</u>, <u>9</u>	7, <u>9</u>
$B^{15}A^4$	6, <u>9</u>	7, <u>9</u>
$B^{15}B^4$	8, 0	<u>8</u>, <u>11</u>

Player 1 best-replies $A^{15}B^4$ to C (11.5) and $B^{15}B^4$ to D (8). Player 2 is indifferent between C and D (9 vs. 9) against $A^{15}B^4$ and against $B^{15}A^4$, and strictly prefers C against $A^{15}A^4$ and D against $B^{15}B^4$. Cells where both payoffs are underlined: $(A^{15}B^4, C)$ and $(B^{15}B^4, D)$.

Interim check. In $(A^{15}B^4, C)$: type 15 plays A ($15 > 8$ against C), type 4 plays B ($8 > 4$), and player 2 earns 9 from either action. In $(B^{15}B^4, D)$: both types of player 1 play B ($8 > 6$ against D) and player 2 earns $11 > 0$ from D.

(d) Interpretation. The first BNE reproduces the informative case: player 1's action reveals his type, and player 2, who cannot condition on it, happens to be indifferent. The second is the complete-information equilibrium (B, D) that survives for both values of x.

(e) Verification. `check_E1.py` recomputes the eight cells and compares them with the book's Fig. 7.5 ($9.5, 18$; $6, 7$; $11.5, 9$; $7, 9$; $6, 9$; $7, 9$; $8, 0$; $8, 11$), and enumerates all 4×2 profiles: exactly two BNE.

Answer. (a) $x = 15$: $(A, C), (B, D)$; $x = 4$: (B, D). (b) BNE: $(A^{15}B^4, C)$ and $(B^{15}B^4, D)$.

Item 3: Gibbons Problem 3.4

Foundations · difficulty 2/5

Statement (verbatim from the list, item 3)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §3.5 Problems, Section 3.1, Problem 3.4, pp. 169–170 [PDF pp. 175–176]. Reproduced verbatim. **Answer in the source:** no.

3.4. Find all the pure-strategy Bayesian Nash equilibria in the following static Bayesian game:

1. Nature determines whether the payoffs are as in Game 1 or as in Game 2, each game being equally likely.
2. Player 1 learns whether nature has drawn Game 1 or Game 2, but player 2 does not.
3. Player 1 chooses either T or B; player 2 simultaneously chooses either L or R.
4. Payoffs are given by the game drawn by nature.

Game 1	L	R
T	1, 1	0, 0
B	0, 0	0, 0

Game 2	L	R
T	0, 0	0, 0
B	0, 0	2, 2

(a) What is asked. All pure-strategy BNE when Nature picks Game 1 or Game 2 with probability $\frac{1}{2}$, player 1 observes which, player 2 does not.

(b) Method. Player 1's strategy is a pair $(s_1(G_1), s_1(G_2)) \in \{T, B\}^2$; player 2 chooses L or R with belief $\frac{1}{2}$ on each game. Find the type-by-type best replies of player 1, then player 2's expected payoffs from L and R (Gibbons §3.1.B, pp. 149–151).

(c) Solution. **Player 1 (each type).** In Game 1: against L, T (payoff 1) beats B (0); against R he is indifferent (0 either way). In Game 2: against L he is indifferent $(0, 0)$; against R, B (2) beats T (0).

Player 2. Given s_1, with $\mathbf{1}[\cdot]$ an indicator,

$$EU_2(L) = \tfrac{1}{2}\,\mathbf{1}[s_1(G_1) = T] \cdot 1 + \tfrac{1}{2} \cdot 0, \qquad EU_2(R) = \tfrac{1}{2} \cdot 0 + \tfrac{1}{2}\,\mathbf{1}[s_1(G_2) = B] \cdot 2.$$

So L is a best reply iff $\frac{1}{2}\mathbf{1}[s_1(G_1) = T] \geq \mathbf{1}[s_1(G_2) = B]$, and R iff the reverse.

- *Player 2 plays L.* Player 1 must play T in G_1; in G_2 any action is a best reply to L. L is optimal for 2 only if $\frac{1}{2} \geq \mathbf{1}[s_1(G_2) = B]$, i.e. $s_1(G_2) = T$. Profile (TT, L): $EU_2(L) = \frac{1}{2} > 0 = EU_2(R)$. ✓
- *Player 2 plays R.* Player 1 must play B in G_2; in G_1 any action is a best reply. R is optimal for 2 iff $\mathbf{1}[s_1(G_2) = B] = 1 \geq \frac{1}{2}\mathbf{1}[s_1(G_1) = T]$, which holds. Profiles (TB, R) and (BB, R). ✓

The remaining profiles fail: BT is never a best reply of player 1 (against L it plays B in G_1, strictly worse than T; against R it plays T in G_2, strictly worse than B), and (TB, L) gives $EU_2(L) = \frac{1}{2} < 1 = EU_2(R)$.

(d) Interpretation. Two equilibria are payoff-dominant in different games: (TT, L) pays $(1, 1)$ in G_1 and $(0, 0)$ in G_2, (BB, R) pays $(0, 0)$ in G_1 and $(2, 2)$ in G_2. Player 2's ignorance creates a coordination problem.

(e) Verification. `check_E1.py` enumerates the 4×2 Bayesian normal form: exactly three pure BNE, $\{(TT, L), (TB, R), (BB, R)\}$. Gibbons gives no answer.

Answer. Three pure-strategy BNE: (TT, L), (TB, R), (BB, R), writing player 1's strategy as (action in Game 1, action in Game 2).

Item 4: Tadelis Exercise 12.1

Foundations · difficulty 2/5

Statement (verbatim from the list, item 4)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 12.1 "Chicken Revisited", p. 266 [PDF p. 285]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Chicken Revisited:** Consider the game of chicken in Section 12.2.1 with the parameters $R = 8$, $H = 16$, and $L = 0$ as described there. A preacher, who knows some game theory, decides to use this model to claim that moving to a society in which all parents are lenient will have detrimental effects on the behavior of teenagers. Does equilibrium analysis support this claim? What if $R = 8$, $H = 0$, and $L = 16$?

Referred to in the statement. The game of chicken of Section 12.2.1 (Tadelis pp. 252–254 [PDF p. 271–273]) (reproduced from the book; Figure 12.3, the game tree, is replaced by the matrix the book derives from it):

> Two teenagers, players 1 and 2, have borrowed their parents' cars and decided to play the game of chicken as follows: They drive toward each other in the middle of a street, and just before impact they must simultaneously choose whether to be chicken and swerve to the right, or continue driving head on. If both are chicken then both gain no respect from their friends but suffer no losses; thus both get a payoff of 0. If i continues to drive while $j \neq i$ plays chicken then i gains all the respect, which is a payoff of R, and j gets no respect, which is worth 0. In this case both players suffer no additional losses. Finally if both continue to drive head on then they split the respect (because respect is considered to be relative), but an accident is bound to happen and they will at least be reprimanded by their parents, if not seriously injured. An accident imposes a personal loss of k on each player, so the payoff to each one is $\frac{R}{2} - k$.
>
> There is, however, a potential difference between these two youngsters: The punishment, k, depends on the type of parents they have. For each kid, parents can be either harsh (H) or lenient (L) with equal probability, and the draws from Nature on the types of parents are independently distributed. This means that the likelihood that player i's parents are harsh is equal to $\frac{1}{2}$ and is independent of the type of parents that player j has. If player i's parents are harsh then they will beat the living daylights out of their child, which imposes a high cost of an accident, denoted by $k = H$. If instead the parents are lenient then they will give their child a long lecture on why his behavior is unacceptable, which imposes a lower cost of an accident, denoted by $k = L < H$. Each kid knows the type of his parents but does not know the type of his opponent's parents. The distribution of types is common knowledge (this is the common prior assumption).
>
> [...] The action sets of each player are $A_1 = \{C, D\}$ and $A_2 = \{c, d\}$, where C (or c) stands for "chicken" and D (or d) stands for "drive."
>
> From the extensive form in Figure 12.3 we can derive the matrix form as follows. A strategy for player 1 is denoted by $xy \in S_1 = \{CC, CD, DC, DD\}$, where x is what he does if he is an L type and y is what he does if he is an H type. Similarly $S_2 = \{cc, cd, dc, dd\}$. [...] This results in the following matrix-form Bayesian game:

Player 1 \ Player 2	cc	cd	dc	dd
CC	$0, 0$	$0, \frac{R}{2}$	$0, \frac{R}{2}$	$0, R$
CD	$\frac{R}{2}, 0$	$\frac{3R}{8} - \frac{H}{4}, \frac{3R}{8} - \frac{H}{4}$	$\frac{3R}{8} - \frac{H}{4}, \frac{3R}{8} - \frac{L}{4}$	$\frac{R}{4} - \frac{H}{2}, \frac{3R}{4} - \frac{L}{4} - \frac{H}{4}$
DC	$\frac{R}{2}, 0$	$\frac{3R}{8} - \frac{L}{4}, \frac{3R}{8} - \frac{H}{4}$	$\frac{3R}{8} - \frac{L}{4}, \frac{3R}{8} - \frac{L}{4}$	$\frac{R}{4} - \frac{L}{2}, \frac{3R}{4} - \frac{L}{4} - \frac{H}{4}$
DD	$R, 0$	$\frac{3R}{4} - \frac{L}{4} - \frac{H}{4}, \frac{R}{4} - \frac{H}{2}$	$\frac{3R}{4} - \frac{L}{4} - \frac{H}{4}, \frac{R}{4} - \frac{L}{2}$	$\frac{R}{2} - \frac{L}{2} - \frac{H}{2}, \frac{R}{2} - \frac{L}{2} - \frac{H}{2}$

> To solve for the Bayesian Nash equilibria we need to have more information about the parameters R, H, and L. Assume that $R = 8$, $H = 16$, and $L = 0$, which results in the following matrix-form game:

Player 1 \ Player 2	cc	cd	dc	dd
CC	$0,0$	$0,4$	$0,4$	$0,8$
CD	$4,0$	$-1,-1$	$-1,3$	$-6,2$
DC	$4,0$	$3,-1$	$3,3$	$2,2$
DD	$8,0$	$2,-6$	$2,2$	$-4,-4$

A quick analysis reveals that the game has a unique pure-strategy Bayesian Nash equilibrium: (DC, dc).

(a) What is asked. Re-solve the chicken game of §12.2.1 for $(R, H, L) = (8, 16, 0)$ and for $(8, 0, 16)$, and say whether equilibrium analysis supports the claim that a society of lenient parents makes teenagers behave worse.

(b) Method. Each kid has type $k \in \{L, H\}$ (cost of an accident), independent and equiprobable. Payoffs of i: swerve $(C) = 0$; drive (D) against C gives R; D against D gives $R/2 - k$. Strategy xy: x if the kid's parents are lenient (L), y if harsh (H); build the 4×4 Bayesian normal form and underline best replies (Tadelis pp. 252–255).

(c) Solution. **Case** $R = 8, H = 16, L = 0$ (book's matrix, rows CC, CD, DC, DD, columns cc, cd, dc, dd):

	cc	cd	dc	dd
CC	$0,0$	$0,4$	$0,4$	$0,8$
CD	$4,0$	$-1,-1$	$-1,3$	$-6,2$
DC	$4,0$	$3,-1$	$\underline{3},\underline{3}$	$2,2$
DD	$8,0$	$2,-6$	$2,2$	$-4,-4$

The unique pure BNE is (DC, dc): lenient kids $(k = 0)$ drive, harsh kids $(k = 16)$ swerve. *Preacher's claim:* in a society where *all* parents are lenient, both kids have $k = 0$ and D is strictly dominant ($4 > 0$ against D, $8 > 0$ against C): everybody drives and the accident probability is 1, against $\frac{1}{4}$ in the mixed society (only lenient–lenient pairs crash). The analysis supports the claim.

Case $R = 8, H = 0, L = 16$**.** Now lenient parents impose the *high* cost. Matrix:

	cc	cd	dc	dd
CC	$0,0$	$0,4$	$0,4$	$0,8$
CD	$4,0$	$\underline{3},\underline{3}$	$3,-1$	$2,2$
DC	$4,0$	$-1,3$	$-1,-1$	$-6,2$
DD	$8,0$	$2,2$	$2,-6$	$-4,-4$

The unique pure BNE is (CD, cd): lenient kids swerve, harsh $(k = 0)$ kids drive (D is dominant for them). Check for a lenient kid: the rival drives with probability $\frac{1}{2}$ at least, so D yields $\frac{1}{2} \cdot 8 + \frac{1}{2}(4 - 16) = -2 < 0$. The prediction reverses: it is the *harsh*-parent kids who are aggressive. A society of all-lenient parents ($k = 16$ for both) is an ordinary chicken game with a symmetric mixed equilibrium in which each drives with probability q solving $8(1 - q) + q(4 - 16) = 0$, $q = \frac{2}{5}$, so accidents have probability $\frac{4}{25} < \frac{1}{4}$, and the pure asymmetric equilibria have none. The claim is not supported.

(d) Interpretation. The conclusion is driven by the payoff story, not by the equilibrium concept: the same model supports or reverses the preacher depending on whether leniency lowers or raises the perceived cost of an accident. This is exactly the book's own caveat after the first matrix.

(e) Verification. `check_E1.py` computes both 4×4 Bayesian normal forms from the types (first case matches the printed entries $(CD, dd) = (-6, 2)$ and $(DC, dc) = (3, 3)$) and enumerates all pure BNE: one each; it also checks $q = \frac{2}{5}$. Tadelis gives no answer.

Answer. $(R, H, L) = (8, 16, 0)$: unique BNE (DC, dc); the all-lenient society drives for sure, so the claim is supported. $(8, 0, 16)$: unique BNE (CD, cd) (harsh kids drive); the all-lenient society is a chicken game with accident probability $\leq \frac{4}{25}$; the claim is not supported.

Item 5: Tadelis Exercise 13.1

Foundations · difficulty 1/5

Statement (verbatim from the list, item 5)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 13.1 "Second-Price Auctions", p. 285 [PDF p. 304]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Second-Price Auctions:** Show that in a second-price sealed-bid auction bidding your valuation weakly dominates bidding above your valuation.

(a) What is asked. Show that bidding one's valuation weakly dominates bidding above it in a second-price sealed-bid auction.

(b) Method. Fix the highest rival bid r (any number) and compare payoffs state by state (Tadelis §13.1.1, pp. 271–273). Bidder i has value v and bids b; if he wins he pays r.

(c) Solution. Let $b > v$. Payoff from bidding v: $v - r$ if $r < v$, 0 if $r > v$ (a tie at $r = v$ pays $v - r = 0$). Payoff from bidding b: $v - r$ if $r < b$, 0 if $r > b$.

rival bid r	bid v	bid $b > v$
$r < v$	$v - r > 0$	$v - r$ (same)
$v \le r < b$	0	$v - r \le 0$ (wins and pays $r \ge v$)
$r > b$	0	0 (same)

In no state does b do better than v; in the middle state ($v < r < b$) it does strictly worse ($v - r < 0$). So $b = v$ weakly dominates every $b > v$. (Tie at $r = b$ pays $v - r \le 0$ in expectation: no gain either.)

(d) Interpretation. Overbidding only adds the states where the bidder wins at a price above his value; the price is the rival's bid, not his own, so he gains nothing in the states where he already won. Underbidding is dominated by the symmetric argument (Muñoz-Garcia item 6).

(e) Verification. `check_E1.py` checks on a grid of 13 values, 13 bids and 13 rival bids (fair coin at ties) that $b = v$ is never worse than any b, and strictly better for some r whenever $b \neq v$. Tadelis gives no answer.

Answer. For every highest rival bid r, bidding v earns at least what any $b > v$ earns, and strictly more when $v < r < b$. Hence bidding the valuation weakly dominates bidding above it.

Item 6: Muñoz-Garcia Ch. 8, Exercise 4

Foundations · difficulty 2/5

Statement (verbatim from the list, item 6)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 8, Exercise 4 "Second Price Auction", p. 362 [PDF p. 372]. Reproduced verbatim (statement only; the book's worked answer follows it). **Answer in the source:** yes, the book's worked answer follows the exercise.

> Consider a setting where every bidder privately observes his/her own valuation for the object. In addition, assume that N bidders interact in an auction in which they simultaneously and independently submit their own bids, the highest bidder wins the object, but he/she pays the second-highest bid (this is the so-called second-price auction, SPA). All other bidders pay zero.
>
> a. Argue that it is a weakly dominant strategy for a bidder to bid her value in the SPA.
>
> b. Are optimal bids affected if the joint distribution of bidders' values exhibits correlation, i.e., if bidder i observes that he has a high valuation for the object he can infer that other bidders must also have a high valuation (in the case of positively correlated values) or that, instead, they must have a low valuation for the object (in the case of negatively correlated values)?
>
> c. Are optimal bids affected by the number of bidders participating in the auction? Are optimal bids affected by bidders' risk aversion?

(a) What is asked. (a) Bidding one's value is weakly dominant in the second-price auction with N bidders. (b) Whether correlated values change optimal bids. (c) Whether N or risk aversion changes them.

(b) Method. Let $r = \max_{j \neq i} b_j$. A weakly dominant strategy is optimal against *every* profile of rivals, hence against every belief about them (Tadelis §13.1.1, pp. 271–273).

(c) Solution. **(a)** Compare $b_i = v_i$ with $b_i \neq v_i$ for each r. If $b_i > v_i$: for $r < v_i$ both win at price r; for $v_i < r < b_i$ the deviation wins and earns $v_i - r < 0$ against 0; for $r > b_i$ both lose. If $b_i < v_i$: for $r < b_i$ both win at r; for $b_i < r < v_i$ the deviation loses and earns 0 against $v_i - r > 0$; for $r > v_i$ both lose. A tie at $r = b_i$ pays no more than $v_i - r$ times the winning probability, never more than bidding v_i. So $b_i = v_i$ is never worse and is strictly better in one state: weak dominance.

(b) No. Correlation changes i's belief about r, but dominance holds state by state, so it holds under any joint distribution of (v_{-i}, b_{-i}): positive or negative correlation, or any N-dimensional dependence, leave $b_i = v_i$ optimal.

(c) No, to both. N enters only through the distribution of r, which is irrelevant for a dominant strategy. Risk aversion: for any strictly increasing utility u with $u(0) = 0$, the state-by-state ranking of $u(v_i - r)$ vs. $u(0)$ is the same as for money, since $v_i - r \gtrless 0$ decides each comparison. The bidding function is the 45° line $b_i(v_i) = v_i$ whatever N and preferences.

(d) Interpretation. Contrast with the first-price auction (items 7, 15): there the bid trades off winning probability against the price paid, so it depends on N, the value distribution and risk attitude. In the second-price auction the price paid does not depend on one's own bid.

(e) Verification. `check_E1.py`: (i) pointwise dominance on a 13^3 grid for three utilities (linear, concave, convex with $u(0) = 0$); (ii) for $N = 2, 3, 5$ and 200 arbitrary joint laws of rival bids, expected payoff from $b = v$ is at least that from every grid bid. M&R's printed answer is the same on all three parts.

Answer. (a) $b_i = v_i$ is weakly dominant. (b) No: dominance is independent of how rivals' values are distributed or correlated. (c) No: neither N nor risk aversion affects the optimal bid.

Item 7: Muñoz-Garcia Ch. 8, Exercise 1

Foundations · difficulty 2/5

Statement (verbatim from the list, item 7)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 8, Exercise 1 "First Price Auction with Uniformly Distributed Valuations—Two Players", p. 350 [PDF p. 360]. Reproduced verbatim (statement only; the book's worked answer follows it). **Answer in the source:** yes, the book's worked answer follows the exercise.

> Consider a first-price auction with two players, each of them drawing his valuation for the object v_i from a uniform distribution $v_i \sim U[0,1]$. Every player i simultaneously and independently submits his bid for the object, b_i, and the seller chooses the player who submitted the highest bid as the winning bidder, who pays the bid he submitted. The other bidder loses the auction and receives a zero payoff.
>
> (a) Set up every bidder i's expected utility maximization problem. Interpret.
>
> (b) Find the equilibrium bidding function that every bidder i uses in the Bayesian Nash Equilibrium (BNE) of the first-price auction.

(a) What is asked. First-price auction, two bidders, $v_i \sim U[0,1]$ independent: (a) set up bidder i's expected-utility problem; (b) find the symmetric BNE bidding function.

(b) Method. Look for $b_i(v_i) = \alpha v_i$ with $\alpha \in (0,1)$ (Gibbons §3.2.B, pp. 155–158; Tadelis §13.1.3, pp. 276–279). A bid x wins iff the rival's bid is below x.

(c) Solution. **(a)** If i bids x and the rival uses $b_j = \alpha v_j$, then $\Pr(\text{win}) = \Pr(\alpha v_j < x) = \Pr(v_j < x/\alpha) = x/\alpha$ (for $x \le \alpha$). So

$$EU_i(x \mid v_i) = \underbrace{\frac{x}{\alpha}}_{\Pr(\text{win})}(v_i - x) + \left(1 - \frac{x}{\alpha}\right)\cdot 0 = \frac{x\,(v_i - x)}{\alpha}.$$

Interpretation: bidding more raises the probability of winning but lowers the margin $v_i - x$ earned on a win; bids above v_i give a negative margin, so $\alpha < 1$ (bid shading).

(b) First-order condition: $\dfrac{\partial EU_i}{\partial x} = \dfrac{v_i - 2x}{\alpha} = 0 \Rightarrow x = \dfrac{v_i}{2}$; second derivative $-2/\alpha < 0$. Consistency with the assumed form $x = \alpha v_i$ requires $\alpha = \frac{1}{2}$. Hence

$$b_i^*(v_i) = \tfrac{1}{2}\,v_i, \qquad EU_i^*(v_i) = \frac{(v_i/2)(v_i/2)}{1/2} = \frac{v_i^2}{2}.$$

Example: $v_i = \frac{2}{3}$ gives the bid $\frac{1}{3}$, as in the book.

Revenue. The winner's bid is $\frac{1}{2}\max(v_1, v_2)$; $E[\max] = \frac{2}{3}$, so expected revenue is $\frac{1}{3}$, equal to the second-price revenue $E[\min(v_1, v_2)] = \frac{1}{3}$ (revenue equivalence).

(d) Interpretation. Half of the value is "paid" for the competition of a single rival; with n bidders the shading falls to v/n (item 15).

(e) Verification. `check_E1.py`: sympy solves the FOC ($x = v/2$, then $\alpha = \frac{1}{2}$); scipy maximizes $\min(2x, 1)(v - x)$ numerically for five values of v and returns $v/2$ with value $v^2/2$; sympy integrates both revenues to $\frac{1}{3}$. Matches M&R's printed $x(v_i) = \frac{1}{2}v_i$.

Answer. (a) $EU_i(x \mid v_i) = \frac{x}{\alpha}(v_i - x)$ against the rival's $b_j = \alpha v_j$. (b) $b_i^*(v_i) = v_i/2$ for both bidders.

Item 8: Tadelis Exercise 13.2

Foundations · difficulty 2/5

Statement (verbatim from the list, item 8)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 13.2 "Complete Information", p. 285 [PDF p. 304]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Complete Information:** Consider a set of N players participating in a sealed-bid second-price auction, but assume that there is complete information so that each player knows the valuation of every other player.
>
> a. Is it still true that each player bidding his valuation is a weakly dominant strategy?
>
> b. Are there other Nash equilibria of this game?

(a) What is asked. N bidders in a second-price auction with complete information: (a) is bidding one's value still weakly dominant? (b) are there other Nash equilibria?

(b) Method. (a) The dominance argument of item 5 never used the others' information. (b) Look for profiles where the winner is not the highest-value bidder, using that the winner pays the second-highest bid (Tadelis §13.1.1).

(c) Solution. **(a) Yes.** The state-by-state comparison with the highest rival bid r holds for every r whatever the bidders know, so $b_i = v_i$ is weakly dominant (items 5–6); in particular $(v_1, \ldots, v_N)$ is a Nash equilibrium. With $v_1 > v_2 \geq \ldots$, bidder 1 wins and pays v_2.

(b) Yes, many. Take any bidder k and let $b_k = \max_{j \neq k} v_j$, $b_j = 0$ for $j \neq k$ ($N \geq 2$). Bidder k wins and pays 0, earning $v_k \geq 0$; he cannot do better and has no reason to deviate. Any other bidder j can win only by bidding $> b_k$ and then pays $b_k = \max_{l \neq k} v_l \geq v_j$, earning $v_j - b_k \leq 0$. So no one deviates: a Nash equilibrium in which bidder k wins even if his value is lowest. Example, $v = (3, 2, 1)$: the profile $(0, 3, 0)$ is an equilibrium with the value-2 bidder winning at price 0; so is $(0, 0, 3)$ with the value-1 bidder winning.

(d) Interpretation. These equilibria use weakly dominated strategies ($b_k > v_k$ is weakly dominated by $b_k = v_k$) and rest on the other bidders' staying out; under complete information nothing in Nash equilibrium rules them out. Eliminating weakly dominated strategies selects the truthful equilibrium, as in item 5.

(e) Verification. `check_E1.py` enumerates all profiles on the integer bid grid $\{0, \ldots, 4\}^3$ for $v = (3, 2, 1)$ (fair coin at ties): the truthful profile and $(0, 3, 0)$, $(0, 0, 3)$ are Nash; 55 pure equilibria in all, in which every bidder wins in some equilibrium; dominance of $b_i = v_i$ is verified against every pair of rival bids. Tadelis gives no answer.

Answer. (a) Yes, bidding the value is still weakly dominant. (b) Yes: e.g. one bidder bids the highest rival value and everyone else bids 0; that bidder wins at price 0, whoever he is.

Item 9: Tadelis Exercise 12.2

Exam · difficulty 3/5

Statement (verbatim from the list, item 9)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 12.2 "Cournot Revisited", p. 266 [PDF p. 285]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Cournot Revisited:** Consider the Cournot duopoly model in which two firms, 1 and 2, simultaneously choose the quantities they supply, q_1 and q_2. The price each will face is determined by the market demand function $p(q_1, q_2) = a - b(q_1 + q_2)$. Each firm has a probability μ of having a marginal unit cost of c_L and a probability $1 - \mu$ of having a marginal unit cost of c_H. These probabilities are common knowledge, but the true type is revealed only to each firm individually. Solve for the Bayesian Nash equilibrium.

(a) What is asked. The BNE of the Cournot duopoly $p = a - b(q_1 + q_2)$ in which each firm independently has cost c_L with probability μ and $c_H > c_L$ with probability $1 - \mu$, privately known.

(b) Method. Types $T_i = \{c_L, c_H\}$, actions $q_i \geq 0$, beliefs $p_i(c_j \mid c_i) = \mu$ for c_L (independent types). A strategy is a pair $(q_i(c_L), q_i(c_H))$. Write the FOC of each type, given the rival's *average* output $\bar{q}_j = \mu q_j(c_L) + (1 - \mu) q_j(c_H)$ (Gibbons §3.1.A, pp. 144–149; Tadelis §12.2.2). Assume interior solutions.

(c) Solution. A firm of type c maximizes $E[(a - b(q + q_j) - c)\, q] = (a - c - b\bar{q}_j - bq)\, q$: $a - c - b\bar{q}_j - 2bq = 0$, i.e.

$$q_i(c) = \frac{a - c - b\bar{q}_j}{2b}, \qquad \text{concave since } -2b < 0.$$

The two types of each firm give the system ($\bar{c} = \mu c_L + (1 - \mu) c_H$)

$$q_L = \frac{a - c_L - b\bar{q}}{2b}, \quad q_H = \frac{a - c_H - b\bar{q}}{2b}, \quad \bar{q} = \mu q_L + (1 - \mu) q_H.$$

Averaging the first two: $2b\bar{q} = a - \bar{c} - b\bar{q}$, so $\bar{q} = \dfrac{a - \bar{c}}{3b}$. Substituting,

$$q_L^* = \frac{2a - 3c_L + \bar{c}}{6b} = \frac{a - c_L}{3b} + \frac{(1 - \mu)(c_H - c_L)}{6b}, \qquad q_H^* = \frac{2a - 3c_H + \bar{c}}{6b} = \frac{a - c_H}{3b} - \frac{\mu(c_H - c_L)}{6b}.$$

The linear system for the four unknowns (two firms, two types) has a unique solution, so the equilibrium is symmetric and unique: both firms use (q_L^*, q_H^*).

(d) Interpretation. Complete-information Cournot gives $(a - c)/(3b)$ for a firm facing a rival with the same cost. Here the low-cost type produces *more* than that when the rival may be high-cost ($1 - \mu > 0$) and the high-cost type produces *less* when the rival may be low-cost: each type adjusts to the rival's expected output.

(e) Verification. `check_E2.py`: sympy solves both the 2-equation and the full 4-equation system and confirms the formulas and $\bar{q} = (a - \bar{c})/(3b)$; scipy maximizes expected profit against the equilibrium rival for $(a, b, c_L, c_H, \mu) = (10, 1, 1, 3, \frac{2}{5})$ and returns q_L^*, q_H^*. Tadelis gives no answer.

Answer. $q^*(c_L) = \dfrac{a - c_L}{3b} + \dfrac{(1 - \mu)(c_H - c_L)}{6b}$ and $q^*(c_H) = \dfrac{a - c_H}{3b} - \dfrac{\mu(c_H - c_L)}{6b}$ for both firms (interior case).

Item 10: Gibbons Problem 3.3

Exam · difficulty 3/5

Statement (verbatim from the list, item 10)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §3.5 Problems, Section 3.1, Problem 3.3, p. 169 [PDF p. 175]. Reproduced verbatim. The demand function is printed with a minus sign before $b_i \cdot p_j$; the scanned copy in `Livros/` carries a reader's handwritten mark over that sign. **Answer in the source:** no.

> **3.3.** Consider the following asymmetric-information model of Bertrand duopoly with differentiated products. Demand for firm i is $q_i(p_i, p_j) = a - p_i - b_i \cdot p_j$. Costs are zero for both firms. The sensitivity of firm i's demand to firm j's price is either high or low. That is, b_i is either b_H or b_L, where $b_H > b_L > 0$. For each firm, $b_i = b_H$ with probability θ and $b_i = b_L$ with probability $1 - \theta$, independent of the realization of b_j. Each firm knows its own b_i but not its competitor's. All of this is common knowledge. What are the action spaces, type spaces, beliefs, and utility functions in this game? What are the strategy spaces? What conditions define a symmetric pure-strategy Bayesian Nash equilibrium of this game? Solve for such an equilibrium.

(a) What is asked. Spell out the Bayesian game (actions, types, beliefs, payoffs, strategies) of Bertrand duopoly with demand $q_i = a - p_i - b_i p_j$, $b_i \in \{b_H, b_L\}$ private, write the symmetric pure BNE conditions and solve.

(b) Method. Gibbons §3.1.A (pp. 144–149). The type is the demand sensitivity b_i; firm i's profit is $\pi_i = p_i\,(a - p_i - b_i p_j)$ (zero cost).

(c) Solution. **The game.** Players $i = 1, 2$. Actions $A_i = [0, \infty)$ (prices). Types $T_i = \{b_H, b_L\}$. Beliefs $p_i(b_j \mid b_i) = \theta$ for $b_j = b_H$ and $1 - \theta$ for b_L, independent of b_i. Payoffs $u_i(p_i, p_j; b_i) = p_i(a - p_i - b_i p_j)$. Strategies $p_i : T_i \to [0, \infty)$, i.e. a pair $(p_i(b_H), p_i(b_L))$.

BNE conditions. A symmetric pure profile (p_H, p_L) (same function for both firms) is a BNE iff for each $b \in \{b_H, b_L\}$

$$p_b \in \arg\max_p \ \theta\, p\,(a - p - b\,p_H) + (1 - \theta)\, p\,(a - p - b\,p_L) = \arg\max_p \ p\,(a - p - b\,\bar{p}),$$

where $\bar{p} = \theta p_H + (1 - \theta) p_L$ (the rival's expected price; profit is linear in p_j).

Solution. FOC $a - 2p - b\bar{p} = 0$ (strictly concave: $-2 < 0$), so $p_b = (a - b\bar{p})/2$. Averaging over types with weights $\theta, 1 - \theta$: $\bar{p} = \dfrac{a - \bar{b}\bar{p}}{2}$, where $\bar{b} = \theta b_H + (1 - \theta) b_L$, so $\bar{p} = \dfrac{a}{2 + \bar{b}}$ and

$$p_H^* = \frac{a}{2}\Big(1 - \frac{b_H}{2 + \bar{b}}\Big) = \frac{a\,(2 + \bar{b} - b_H)}{2\,(2 + \bar{b})}, \qquad p_L^* = \frac{a\,(2 + \bar{b} - b_L)}{2\,(2 + \bar{b})}.$$

Since $b_H > b_L$, $p_H^* < p_L^*$: the type whose demand reacts more to the rival's price prices lower. Positivity requires $2 + \bar{b} > b_H$, true e.g. for $b_H < 2$.

(d) Interpretation. With $b_H = b_L = b$ the price is $a/(2 + b)$, the standard differentiated-products Bertrand price. Here the rival's type is unknown, so each firm prices against the *expected* rival price.

(e) Verification. `check_E2.py`: sympy solves the two FOCs for (p_H, p_L) and matches the formulas and $\bar{p} = a/(2 + \bar{b})$; scipy maximizes profit for $(a, b_H, b_L, \theta) = (10, \frac{3}{2}, \frac{1}{2}, \frac{1}{3})$ against the equilibrium rival and returns p_H^*, p_L^*. Gibbons gives no answer.

Answer. Symmetric BNE: $p^*(b) = \dfrac{a\,(2+\bar{b}-b)}{2\,(2+\bar{b})}$ with $\bar{b} = \theta b_H + (1-\theta) b_L$, for $b \in \{b_H, b_L\}$.

Item 11: Tadelis Exercise 12.3

Exam · difficulty 3/5

Statement (verbatim from the list, item 11)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 12.3 "Armed Conflict", pp. 266–267 [PDF pp. 285–286]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Armed Conflict:** Consider the following strategic situation: Two rival armies plan to seize a disputed territory. Each army's general can choose either to attack (A) or to not attack (N). In addition, each army is either strong (S) or weak (W) with equal probability, and the realizations for each army are *independent*. Furthermore the type of each army is known only to that army's general. An army can capture the territory if either (i) it attacks and its rival does not or (ii) it and its rival attack, but it is strong and the rival is weak. If both attack and are of equal strength then neither captures the territory. As for payoffs, the territory is worth m if captured and each army has a cost of fighting equal to s if it is strong and w if it is weak, where $s < w$. If an army attacks but its rival does not, no costs are borne by either side. Identify all the pure-strategy Bayesian Nash equilibria of this game for the following two cases, and briefly describe the intuition for your results:
>
> a. $m = 3$, $w = 2$, $s = 1$.
> b. $m = 3$, $w = 4$, $s = 2$.

(a) What is asked. All pure-strategy BNE of the armed-conflict game for (a) $m = 3, w = 2, s = 1$ and (b) $m = 3, w = 4, s = 2$.

(b) Method. Types S or W (equiprobable, independent), actions A or N. An army's payoff from N is 0. From A: m if the rival plays N; if both attack, $m - c$ when i is strong and the rival weak, and $-c$ otherwise, where $c = s$ (strong) or w (weak). A strategy is (x_S, x_W), the action of the strong and of the weak type. For each type compute the expected payoff of A against each of the four rival strategies and compare with 0.

(c) Solution. Expected payoff of A for type t against a rival strategy (y_S, y_W) is $\frac{1}{2}u(A \mid y_S \text{ vs } S) + \frac{1}{2}u(A \mid y_W \text{ vs } W)$; ($N$ yields 0).

(a) $m = 3, w = 2, s = 1$**.**

rival (y_S, y_W)	AA	AN	NA	NN
$E[A]$, strong	$\frac{1}{2}$	1	$\frac{5}{2}$	3
$E[A]$, weak	-2	$\frac{1}{2}$	$\frac{1}{2}$	3

The strong type attacks against every rival strategy ($E[A] > 0$). The weak type attacks unless the rival is AA (always attacks). So a rival AA forces (A, N) from the other army, but against AN the weak type wants A: no symmetric pure equilibrium. The BNE are asymmetric:

$$(AA,\ AN) \quad \text{and} \quad (AN,\ AA).$$

Check (AA, AN): army 1 faces AN and attacks with both types ($1 > 0$, $\frac{1}{2} > 0$); army 2 faces AA, so its strong type attacks ($\frac{1}{2} > 0$) and its weak type stays ($-2 < 0$).

(b) $m = 3, w = 4, s = 2$**.**

rival (y_S, y_W)	AA	AN	NA	NN
$E[A]$, strong	$-\frac{1}{2}$	$\frac{1}{2}$	2	3
$E[A]$, weak	-4	$-\frac{1}{2}$	$-\frac{1}{2}$	3

The weak type attacks only if the rival never attacks ($E[A] = 3$ against NN, negative otherwise); the strong type attacks unless the rival is AA. Profiles:

$$(AN,\, AN), \qquad (AA,\, NN), \qquad (NN,\, AA).$$

Check (AN, AN): strong attacks ($\frac{1}{2} > 0$ against AN), weak stays ($-\frac{1}{2} < 0$). (AA, NN): army 1 attacks with both types because army 2 never does ($3 > 0$), and army 2 stays with both types because army 1 always attacks ($-\frac{1}{2} < 0$ and $-4 < 0$).

(d) Interpretation. (a) The costs are low enough that attacking pays unless the rival is sure to attack. The only equilibria have one army whose weak type attacks and another whose weak type backs off. (b) With high costs of fighting, only the strong type fights; the weak type attacks only against a rival who never does, which supports the lopsided equilibria (AA, NN) and (NN, AA).

(e) Verification. `check_E2.py` computes both Bayesian normal forms from the payoff rules and enumerates all 4×4 pure profiles; it prints the two tables of expected attack payoffs above and asserts exactly the two profiles in (a) and the three in (b). Tadelis gives no answer.

Answer. (a) (AA, AN) and (AN, AA). (b) (AN, AN), (AA, NN) and (NN, AA); each strategy lists the action of the strong type, then of the weak type.

Item 12: Tadelis Exercise 12.9

Exam · difficulty 3/5

Statement (verbatim from the list, item 12)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 12.9 "Grab the Dollar", p. 269 [PDF p. 288]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Grab the Dollar:** Each of two players has two possible actions: grab the dollar or don't grab the dollar. Player i's payoff is 1 if he is the only one to grab the dollar, and his payoff is 0 if he does not. Each player's payoff is -1 if both players grab the dollar.
>
> a. Find the Nash equilibria of this game of complete information.
>
> b. Consider the following perturbation of the game. The payoff structure is the same, with the following addition: When player i is the only one to grab the dollar his payoff is $1+\theta_i$, where θ_i is player i's type and it is uniformly distributed over $[-\varepsilon, \varepsilon]$ with $\varepsilon < 1$. Each player knows only his own type, and the distribution of types is common knowledge. Find the pure-strategy Bayesian Nash equilibrium of this game of incomplete information.
>
> c. Show that when ε converges to 0, the pure-strategy Bayesian Nash equilibrium converges to the mixed-strategy Nash equilibrium of the game with complete information.

(a) What is asked. (a) Nash equilibria of grab-the-dollar. (b) The pure-strategy BNE when $\theta_i \sim U[-\varepsilon, \varepsilon]$ is added to the payoff of the sole grabber. (c) Its limit as $\varepsilon \to 0$.

(b) Method. (a) Best replies; indifference gives the mixed equilibrium. (b) A player grabs when the expected payoff from grabbing is nonnegative, which is increasing in θ_i: a threshold rule (Tadelis §12.5, pp. 264–266, Harsanyi's purification).

(c) Solution. **(a)** Payoffs: sole grabber 1, non-grabber 0, both grab -1 each. Pure Nash equilibria: (G, D) and (D, G). Mixed: if the rival grabs with probability p, grabbing yields $(1-p)\cdot 1 + p\cdot(-1) = 1-2p$ against 0 from not grabbing; indifference gives $p = \frac{1}{2}$. So the mixed equilibrium has both players grabbing with probability $\frac{1}{2}$.

(b) Let the rival grab with probability p. Player i of type θ_i gets $(1-p)(1+\theta_i) - p$ from G and 0 from D, so he grabs iff

$$\theta_i \;\geq\; \frac{2p-1}{1-p}.$$

Take thresholds t_i (i grabs iff $\theta_i \geq t_i$): the rival grabs with probability $p_j = \dfrac{\varepsilon - t_j}{2\varepsilon}$, hence $1 - p_j = \dfrac{\varepsilon + t_j}{2\varepsilon}$, $2p_j - 1 = -t_j/\varepsilon$ and

$$t_i = \frac{-2\,t_j}{\varepsilon + t_j}, \qquad t_j = \frac{-2\,t_i}{\varepsilon + t_i}.$$

Subtracting the two cross-multiplied equations $t_i(\varepsilon+t_j) = -2t_j$ and $t_j(\varepsilon+t_i) = -2t_i$ gives $(\varepsilon-2)(t_i-t_j) = 0$, so $t_i = t_j = t$ because $\varepsilon < 1 < 2$; then $t(\varepsilon + t) = -2t$, i.e. $t = 0$ or $t = -(2+\varepsilon)$, and only $t = 0$ lies in $[-\varepsilon, \varepsilon]$. The interior pure BNE is: *each player grabs iff* $\theta_i \geq 0$, i.e. with probability $\frac{1}{2}$. (Two further pure BNE sit at the corners: one player always grabs, since $1 + \theta \geq 1 - \varepsilon > 0$ when the other never grabs, and the other never grabs, since grabbing against a sure grabber yields -1. They are the perturbed versions of (G, D) and (D, G).)

(c) In the symmetric BNE, the probability of grabbing is $\frac{1}{2}$ for *every* ε, so as $\varepsilon \to 0$ it converges to the mixed-strategy equilibrium of (a), now interpreted as a pure strategy of a slightly uncertain game: players randomize only in the eyes of an observer who does not know their θ_i.

(d) Interpretation. The mixed equilibrium becomes the limit of pure-strategy BNE, with no player randomizing: this is Harsanyi's purification argument.

(e) Verification. `check_E2.py`: sympy solves the two cross equations and returns only $(t_1, t_2) = (0, 0)$ in $[-\varepsilon, \varepsilon]$; the grab probability at $t = 0$ equals $\frac{1}{2}$; the mixed $p = \frac{1}{2}$ is checked. Tadelis gives no answer.

Answer. (a) (G, D), (D, G) and the mixed equilibrium with $p = \frac{1}{2}$. (b) Symmetric BNE: grab iff $\theta_i \geq 0$ (probability $\frac{1}{2}$); there are also the two corner pure BNE. (c) The grabbing probability is $\frac{1}{2}$ for all ε, hence converges to the mixed equilibrium of (a).

Item 13: Tadelis Exercise 12.5

Exam · difficulty 3/5

Statement (verbatim from the list, item 13)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 12.5 "Not All That Glitters", pp. 267–268 [PDF pp. 286–287]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Not All That Glitters:** A prospector owns a gold mine where he can dig to recover gold. His output depends on the amount of gold in the mine, denoted by x. The prospector knows the value of x, but the rest of the world knows only that the amount of gold is uniformly distributed on the interval $[0, 1]$. Before deciding to mine, the prospector can try to sell his mine to a large mining company, which is much more efficient in its extraction methods. The prospector can ask the company owner for any price $p \geq 0$, and the owner can reject (R) or accept (A) the offer. If the owner rejects the offer then the prospector is left to mine himself, and his payoff from self-mining is equal to $3x$. If the owner accepts the offer then the prospector's payoff is the price p, while the owner's payoff is given by the net value $4x - p$, and this is common knowledge.
>
> a. Show that for a given price $p \geq 0$ there is a threshold type $x(p) \in [0, 1]$ of prospector, such that types below $x(p)$ will prefer to sell the mine, while types above $x(p)$ will prefer to self-mine.
>
> b. Find the pure-strategy Bayesian Nash equilibrium of this game, and show that it is unique. What is the expected payoff of each type of prospector and of the company owner in the equilibrium you derived?

(a) What is asked. (a) For a price p, the prospector's threshold type; (b) the pure BNE, its uniqueness and the expected payoffs.

(b) Method. The prospector knows $x \sim U[0, 1]$; the owner does not. This is adverse selection (Tadelis §12.3, pp. 258–261): the types that sell are the worst ones, which the owner must take into account in his expected payoff. Find the owner's expected net value conditional on acceptance.

(c) Solution. **(a)** A prospector of type x who asks p sells iff the price is at least his self-mining value: $p \geq 3x$, i.e. $x \leq x(p) = \min\{p/3,\, 1\}$. The payoff $3x$ increases in x while the price is type-independent, so types below $x(p)$ prefer to sell and types above prefer to mine.

(b) If the owner accepts price $p \in (0, 3]$, he buys exactly the types $x \leq p/3$, with $E[x \mid x \leq p/3] = p/6$:

$$E[4x - p \mid x \leq p/3] = 4 \cdot \frac{p}{6} - p = -\frac{p}{3} < 0.$$

For $p \geq 3$ all types sell and $E[4x - p] = 2 - p < 0$. So the owner loses on average at any positive price and rejects it. Accepting $p = 0$ yields 0 (indifference), but then every type $x > 0$ strictly prefers asking a rejected price and mining ($3x > 0$).

Uniqueness. Suppose some price $p > 0$ were accepted in equilibrium; the highest accepted price is asked by every type $x < p/3$ (they gain $p - 3x > 0$ from it), so the owner's conditional payoff is $-p/3 < 0$: accepting is not a best reply, a contradiction. Hence in every pure BNE the owner accepts no positive price that is asked, and trade never occurs (up to the null type $x = 0$).

Payoffs. Each prospector type earns $3x$ (expected $\frac{3}{2}$), the owner earns 0. The market breaks down although trade would create surplus $4x - 3x = x > 0$ for every type.

(d) Interpretation. The owner cannot tell good mines from bad: any price high enough to attract the good ones is above their average value to him. Akerlof's lemons logic, with a continuum of types.

(e) Verification. `check_E2.py`: sympy integrates $E[4x - p \mid x \le p/3] = -p/3$, $E[x \mid x \le p/3] = p/6$, the full-sale case $2 - p$ and $E[3x] = \frac{3}{2}$. Tadelis gives no answer.

Answer. (a) $x(p) = \min\{p/3, 1\}$. (b) Unique pure BNE: no trade (the owner rejects every $p > 0$, since $E[4x - p \mid x \le p/3] = -p/3$); payoffs $3x$ to the prospector, 0 to the owner.

Item 14: Tadelis Exercise 12.6

Exam · difficulty 4/5

Statement (verbatim from the list, item 14)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 12.6 "Reap and Weep", p. 268 [PDF p. 287]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Reap and Weep:** A farmer owns some land that he can farm to produce crops. Farming output depends on the talent of the farmer. The farmer knows his talent, but the rest of the world knows only that a farmer's talent is uniformly distributed: $\theta \in [0,1]$. The farmer's payoff from farming his land is equal to his talent θ. Before setting up his farm, the farmer approaches the local manufacturing plant and offers to work on the production line. The farmer can ask the plant owner for any wage $w \geq 0$, and the owner can reject (R) or accept (A) the offer. If the owner rejects the offer then the farmer must return home and settle into his farming. If the owner accepts the offer then the farmer's payoff is the wage w, while the owner's payoff is given by the net value $\frac{3}{2}\theta - w$, and this is common knowledge.
>
> a. Define the set of pure strategies for each player and find the pure-strategy Bayesian Nash equilibria of this game.
>
> b. Averaging over the type of farmer, what are the possible levels of *social surplus* (sum of expected payoffs of the farmer and the owner in their potential relationship) from the equilibrium you derived in (a)?
>
> c. A local policy maker who is advocating for the increase of social surplus is proposing to cut water subsidies to the farmers, which would imply that a farmer of type θ would get a payoff of $\frac{1}{2}\theta$ from farming his land. This policy has no effect on the productivity of manufacturing. Using the criterion of social surplus, can you advocate for this policy maker using equilibrium analysis?

(a) What is asked. (a) Strategies and pure BNE of the farmer–plant game; (b) the social surplus in equilibrium; (c) the effect on surplus of cutting the farming payoff to $\theta/2$.

(b) Method. As in item 13: a farmer of talent $\theta \sim U[0,1]$ asks a wage w; the owner accepts or rejects. Strategies: farmer $w : [0,1] \to [0,\infty)$; owner $\{A, R\}$ as a function of w. The owner's expected net value is computed on the set of types that accept (Tadelis §12.3, pp. 258–261).

(c) Solution. **(a)** A farmer of type θ who asks w takes the job iff $w \geq \theta$ (outside option θ), so types $\theta \leq w$ sell. If the owner accepts $w \in (0,1]$ he employs the types $\theta \leq w$, with $E[\theta \mid \theta \leq w] = w/2$:

$$E\big[\tfrac{3}{2}\theta - w \mid \theta \leq w\big] = \tfrac{3}{4}w - w = -\tfrac{w}{4} < 0; \qquad w \geq 1: \ \tfrac{3}{4} - w < 0.$$

The owner rejects every positive wage; at $w = 0$ he is indifferent, but farmers $\theta > 0$ prefer to farm. As in item 13, an accepted positive wage would be asked by every type $\theta < w$ and would give the owner a loss, so it cannot occur in a BNE. *BNE:* the owner rejects all $w > 0$ (and may accept $w = 0$); every farmer farms (any rejected ask is a best reply).

(b) Surplus = farmer's plus owner's expected payoffs $= E[\theta] + 0 = \frac{1}{2}$. It is the only possible level: no trade occurs in any BNE (the efficient level would be $E[\frac{3}{2}\theta] = \frac{3}{4}$).

(c) Now the farmer's outside option is $\theta/2$, so types $\theta \leq 2w$ accept a wage w. For $w \leq \frac{1}{2}$: $E[\frac{3}{2}\theta - w \mid \theta \leq 2w] = \frac{3}{2}w - w = \frac{w}{2} > 0$; for $w \geq \frac{1}{2}$ all types sell and $E = \frac{3}{4} - w \geq 0$ iff $w \leq \frac{3}{4}$. So the owner accepts any wage up to $\frac{3}{4}$, and a farmer asks the highest accepted wage, getting $\frac{3}{4} > \theta/2$ for all θ. The BNE with trade at a wage w^* accepted by the owner has surplus

$$S(w^*) = \tfrac{1}{4} + 2w^{*2} \ (w^* \leq \tfrac{1}{2}), \qquad S(w^*) = \tfrac{3}{4} \ (\tfrac{1}{2} \leq w^* \leq \tfrac{3}{4}),$$

and the no-trade BNE has surplus $\frac{1}{4}$. The maximum, $\frac{3}{4}$, is reached when everybody works: with the owner's best reply (accept every wage that gives him a nonnegative net value) the farmer asks $w = \frac{3}{4}$ and trade is efficient. Since $\frac{3}{4} > \frac{1}{2}$, the criterion *supports* the policy-maker: cutting the subsidy turns a market that had broken down into one with full trade. Caveats: no-trade remains a BNE with surplus $\frac{1}{4} < \frac{1}{2}$ (and trade at $w^* < \sqrt{2}/4 \approx 0.35$ gives less than $\frac{1}{2}$); and the farmer types $\theta > \frac{3}{4}$ are worse off than before the policy (they now earn $\frac{3}{4} < \theta$).

(d) Interpretation. With outside option $\theta/2$ the average talent of those willing to work at wage w, $E[\theta \mid \theta \leq 2w] = w$, is high enough to cover the wage ($\frac{3}{2}w > w$), so adverse selection no longer blocks trade: a worse outside option can raise total surplus.

(e) Verification. `check_E2.py`: sympy confirms $-w/4$ in (a); $E[\theta] = \frac{1}{2}$; in (c) $w/2$, $\frac{3}{4} - w$ (zero at $w = \frac{3}{4}$), the surplus function $\frac{1}{4} + 2w^2$ (equal to $\frac{3}{4}$ at $w = \frac{1}{2}$), the no-trade surplus $\frac{1}{4}$ and the break-even $w = \sqrt{2}/4$. Tadelis gives no answer.

Answer. (a) No-trade BNE: the owner rejects every $w > 0$ ($E = -w/4$). (b) Surplus $\frac{1}{2}$. (c) With outside option $\theta/2$ the owner accepts wages up to $\frac{3}{4}$, and the best BNE has all farmers work at $w = \frac{3}{4}$ with surplus $\frac{3}{4} > \frac{1}{2}$: the policy can be advocated on the surplus criterion (not every BNE improves, and types $\theta > \frac{3}{4}$ lose).

Item 15: Gibbons Problem 3.6

Exam · difficulty 3/5

Statement (verbatim from the list, item 15)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §3.5 Problems, Section 3.2, Problem 3.6, p. 170 [PDF p. 176]. Reproduced verbatim. **Answer in the source:** no.

> **3.6.** Consider a first-price, sealed-bid auction in which the bidders' valuations are independently and uniformly distributed on $[0, 1]$. Show that if there are n bidders, then the strategy of bidding $(n-1)/n$ times one's valuation is a symmetric Bayesian Nash equilibrium of this auction.

(a) What is asked. Show that with n bidders with values i.i.d. $U[0,1]$ in a first-price sealed-bid auction, bidding $\frac{n-1}{n}v_i$ is a symmetric BNE.

(b) Method. Verification, not derivation: take the rivals' strategy $b_j = \frac{n-1}{n}v_j$ as given and show that it is a best reply for every v_i (Gibbons §3.2.B, pp. 155–158).

(c) Solution. Let $k = \frac{n-1}{n}$. A bid $x \le k$ beats rival j iff $kv_j < x$, i.e. $v_j < x/k$, with probability x/k; it beats all $n-1$ independent rivals with probability $(x/k)^{n-1}$. Hence

$$EU_i(x \mid v_i) = \left(\frac{x}{k}\right)^{n-1}(v_i - x), \qquad 0 \le x \le k.$$

FOC: $(n-1)\dfrac{x^{n-2}}{k^{n-1}}(v_i - x) - \dfrac{x^{n-1}}{k^{n-1}} = 0$, i.e. $(n-1)(v_i - x) = x$, so

$$x = \frac{n-1}{n}\,v_i = kv_i.$$

This is the unique interior critical point and a maximum: on $[0,k]$, $EU_i = c\,x^{n-1}(v_i - x)$ with $c > 0$ is 0 at $x = 0$, increasing up to $x = kv_i$ and decreasing afterwards. Bids $x > k$ win with probability 1 and earn $v_i - x$, decreasing in x and continuous with $EU_i(k)$, so they do no better than $x = k$, which is already beaten by kv_i (as $kv_i \le k$, with equality only at $v_i = 1$). So $b_i(v_i) = kv_i$ is a best reply to the other $n-1$ bidders using it: a symmetric BNE.

(d) Interpretation. Shading is v_i/n: it vanishes as $n \to \infty$. The expected revenue is $E[\max v] \cdot k = \frac{n}{n+1} \cdot \frac{n-1}{n} = \frac{n-1}{n+1}$, the expected second-highest value, i.e. the second-price revenue (revenue equivalence); $n = 2$ gives item 7's $\frac{1}{3}$.

(e) Verification. `check_E2.py`: for $n = 2, \ldots, 7$ sympy solves the FOC and returns $(n-1)v/n$; scipy maximizes EU_i (with the winning-for-sure branch for $x > k$) at $v = 0.7$ and returns $(n-1)\cdot 0.7/n$; sympy integrates the two revenues to $(n-1)/(n+1)$. Gibbons gives no answer.

Answer. With the rivals bidding $\frac{n-1}{n}v_j$, $EU_i = (\frac{nx}{n-1})^{n-1}(v_i - x)$ is maximized at $x = \frac{n-1}{n}v_i$: the profile is a symmetric BNE.

Item 16: Tadelis Exercise 13.9

Exam · difficulty 3/5

Statement (verbatim from the list, item 16)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 13.9 "Private Value Linear Bidding", p. 287 [PDF p. 306]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Private Value Linear Bidding:** Two players participate in a first-price sealed-bid auction in an IPV setting. The value of the good for each player is given by $0.5+\theta_i$, where θ_i is observed only by player i and is uniformly distributed over the interval $[0,1]$.
>
> a. Show that there is a symmetric Bayesian Nash equilibrium in which each bidder i's bidding strategy is of the form $s_i^*(\theta_i) = a + b\theta_i$.
>
> b. What is the expected payoff of a player if his type is θ_i?

(a) What is asked. Two bidders, first-price auction, values $0.5+\theta_i$ with $\theta_i \sim U[0,1]$: (a) a symmetric BNE of the form $s(\theta_i) = a + b\theta_i$; (b) the expected payoff of type θ_i.

(b) Method. Method of undetermined coefficients (Tadelis §13.1.3, pp. 276–280): let the rival bid $a + b\theta_j$, write the bidder's expected payoff for a bid x, take the FOC and match coefficients.

(c) Solution. **(a)** A bid $x \in [a, a+b]$ beats the rival iff $a + b\theta_j < x$, i.e. $\theta_j < (x-a)/b$, with probability $(x-a)/b$. So

$$EU_i(x \mid \theta_i) = \frac{x-a}{b}\big(0.5 + \theta_i - x\big).$$

FOC: $(0.5+\theta_i - x) - (x-a) = 0$, so $x = \frac{1}{2}(0.5+\theta_i+a)$ (concave in x). Matching with $x = a + b\theta_i$ for all θ_i:

$$b = \tfrac{1}{2}, \qquad a = \tfrac{1}{2}(0.5+a) \;\Rightarrow\; a = \tfrac{1}{2}.$$

So $s^*(\theta_i) = \frac{1}{2} + \frac{1}{2}\theta_i$, with bids in $[\frac{1}{2}, 1]$. Bids below $a = \frac{1}{2}$ win with probability 0, and bids above 1 win surely at a worse price than 1; neither helps.

(b) The probability of winning with $x = s^*(\theta_i)$ is $(x-a)/b = \theta_i$, and the margin is $0.5 + \theta_i - \frac{1}{2} - \frac{1}{2}\theta_i = \frac{1}{2}\theta_i$, so

$$EU_i^*(\theta_i) = \theta_i \cdot \tfrac{1}{2}\theta_i = \frac{\theta_i^2}{2}.$$

(d) Interpretation. In value terms $v_i = 0.5 + \theta_i$, the bid is $v_i/2 + \frac{1}{4}$: half the value plus $\frac{1}{4}$, because the lowest value is 0.5, so no one needs to bid below 0.5.

(e) Verification. `check_E2.py`: sympy gets $x = (0.5 + \theta + a)/2$ and the unique $(a,b) = (\frac{1}{2}, \frac{1}{2})$; scipy maximizes EU_i numerically at $\theta \in \{0.2, 0.5, 0.9, 1\}$ and returns $0.5 + 0.5\theta$; the equilibrium payoff $\theta^2/2$ is confirmed symbolically and numerically. Tadelis gives no answer.

Answer. (a) $s^*(\theta_i) = \frac{1}{2} + \frac{1}{2}\theta_i$ ($a = b = \frac{1}{2}$). (b) $EU_i^*(\theta_i) = \theta_i^2/2$.

Item 17: Tadelis Exercise 13.3

Exam · difficulty 3/5

Statement (verbatim from the list, item 17)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 13.3 "Reserve Prices (Easier)", pp. 285–286 [PDF pp. 304–305]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Reserve Prices (Easier):** Consider a seller who must sell a single private value good. There are two potential buyers, each with a valuation that can take on one of three values, $\theta_i \in \{0, 1, 2\}$, each value occurring with an equal probability of $\frac{1}{3}$. The players' values are independently drawn. The seller will offer the good using a second-price sealed-bid auction, but he can set a "reserve price" of $r \geq 0$ that modifies the rules of the auction as follows. If both bids are below r then neither bidder obtains the good and it is destroyed. If both bids are at or above r then the regular auction rules prevail. If only one bid is at or above r then that bidder obtains the good and pays r to the seller.
>
> a. Is it still a weakly dominant strategy for each player to bid his valuation when $r > 0$?
> b. What is the expected revenue of the seller when $r = 0$ (no reserve price)?
> c. What is the expected revenue of the seller when $r = 1$?
> d. What explains the difference between your answers to (b) and (c)?
> e. What is the optimal reserve price r for the seller, and what can you conclude about the value of reserve prices?

(a) What is asked. Second-price auction with reserve price r, two buyers, values $\{0, 1, 2\}$ equiprobable and independent: (a) is truthful bidding still weakly dominant? Expected revenue at (b) $r = 0$, (c) $r = 1$; (d) why they differ; (e) the optimal r.

(b) Method. (a) Compare payoffs state by state as in item 5. (b)–(e) Enumerate the nine value pairs (each probability $\frac{1}{9}$) with bids equal to values (Tadelis §13.1.1 and the reserve-price discussion).

(c) Solution. **(a) Yes.** A winner pays $\max\{r, \text{rival bid}\}$ and wins iff his bid is at least r and beats the rival. Value $v < r$: bidding v loses; any winning bid would pay at least $r > v$, a loss. Value $v \geq r$: the situation is that of item 5 with the effective price $\max\{r, \text{rival bid}\}$, so the same table shows no bid beats $b = v$.

(b) $r = 0$**.** Revenue is the second-highest value, $\min(v_1, v_2)$: 0 in 5 pairs, 1 in 3 pairs, 2 in 1 pair:

$$\tfrac{1}{9}(0 \cdot 5 + 1 \cdot 3 + 2 \cdot 1) = \tfrac{5}{9}.$$

(c) $r = 1$**.** Value-0 buyers do not bid. Pairs: $(0, 0)$ no sale (0); one buyer at least 1 and the other 0 (4 pairs): price $r = 1$; both at least 1: price min: $(1, 1), (1, 2), (2, 1) \to 1$ each, $(2, 2) \to 2$. Revenue $= \frac{1}{9}(4 \cdot 1 + 1 + 1 + 1 + 2) = \frac{9}{9} = 1$.

(d) Compared with $r = 0$, the four pairs with exactly one value-0 buyer now yield 1 instead of 0 ($+\frac{4}{9}$): the other buyer used to pay the second-highest bid, 0, and now pays r. The pair $(0, 0)$ yields 0 in both cases and the pairs with both values at least 1 are unaffected (price min $\geq 1 = r$). Nothing is lost, because $r = 1$ is the lowest positive value: it excludes only buyers who would have paid 0.

(e) For $r \in (0, 1]$: revenue $= \frac{5}{9} + \frac{4}{9}r$ (the $(1, 1), (1, 2), (2, 1), (2, 2)$ pairs give 5/9, the four "one buyer" pairs give r). For $r \in (1, 2]$ only value-2 buyers bid: four pairs at price r and $(2, 2)$ at price 2, revenue $= \frac{4r+2}{9}$, increasing in r; $r > 2$ gives 0. The maximum is at $r = 2$ (the highest value) with revenue $\frac{10}{9}$, twice the revenue of $r = 0$ (strictly above $r = 1$'s 1). Conclusion: a reserve price above the seller's value (here 0) raises revenue: the seller gives up sales that would have been at low prices to raise the price paid by the high types; the optimal reserve is above the lowest values.

(d) Interpretation. A reserve price is a monopoly price inside the auction: raising it above 1 also excludes the value-1 buyers, but the extra revenue from value-2 buyers (r instead of min) outweighs the lost sales, because each value-2 buyer is alone in $\frac{4}{9}$ of the pairs.

(e) Verification. `check_E2.py`: enumerates the nine pairs for r on a fine grid and asserts 5/9, 1, 10/9 at $r = 0, 1, 2$, the two linear pieces for $r \in (0, 1]$ and $(1, 2]$, the global maximum 10/9 at $r = 2$; and dominance of $b = v$ against every rival bid for five values of v and four reserves. Tadelis gives no answer.

Answer. (a) Yes. (b) 5/9. (c) 1. (d) Excluding value-0 buyers lets the seller charge 1 in the four pairs with exactly one such buyer, at no loss elsewhere. (e) The optimal reserve is $r = 2$, with revenue 10/9: reserve prices can raise revenue above the no-reserve auction.

Item 18: Tadelis Exercise 12.8

Exam · difficulty 4/5

Statement (verbatim from the list, item 18)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 12.8 "Jury Voting", pp. 268–269 [PDF pp. 287–288]. Reproduced verbatim. The game it refers to is Tadelis §12.4 (pp. 261–264), summarized in Notas/L07/L07-09-common-values-curses §1. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Jury Voting:** Consider the jury voting game in Section 12.4.
>
> a. Show that both players choosing CC is a Bayesian Nash equilibrium of the two-player game.
> b. Would it be better for the two players if only one of them decided on the fate of the defendant? Why or why not?
> c. Consider a three-player game in which the defendant is convicted if at least two players vote to convict him, and acquitted otherwise (i.e., majority voting determines the defendant's fate). Assume that the informational structure is the same as in the two-player game. Are there conditions in which voting according to one's signal (C if $\theta_i = \theta_G$ and A if $\theta_i = \theta_I$) is a Bayesian Nash equilibrium?

Referred to in the statement. The jury voting game of Section 12.4 (Tadelis pp. 261–262 [PDF p. 280–281]) (reproduced from the book):

> As an example, consider a jury made up of two players (jurors) who must collectively decide whether to acquit (A) or to convict (C) a defendant. The process calls for each player to cast a sealed vote, and the defendant is convicted only if both vote C. The problem is that there is uncertainty about whether the defendant is guilty (G) or innocent (I). The prior probability that the defendant is guilty is given by $q > \frac{1}{2}$ and is common knowledge. Assume that each player cares about making the right decision, so that if the defendant is guilty then each player receives a payoff of 1 from a conviction and 0 from an acquittal. If instead the defendant is innocent then each player receives a payoff of 0 from a conviction and 1 from an acquittal.
>
> [...] Now imagine that things are a bit more complex. Each player i has a different expertise and, when observing the evidence, gets a private signal $\theta_i \in \{\theta_G, \theta_I\}$ that contains valuable information. In particular when the defendant is guilty player i is more likely to receive the signal $\theta_i = \theta_G$ than when the defendant is innocent, and vice versa for $\theta_i = \theta_I$. Assume that the signals are independent: the probability of receiving the signal θ_G when the defendant is guilty is equal to the probability of receiving the signal θ_I when the defendant is innocent, so that
>
> $$\Pr\{\theta_i = \theta_G \mid G\} = \Pr\{\theta_i = \theta_I \mid I\} = p > \tfrac{1}{2} \quad \text{for } i \in \{1, 2\}$$
>
> and
>
> $$\Pr\{\theta_i = \theta_G \mid I\} = \Pr\{\theta_i = \theta_I \mid G\} = 1 - p < \tfrac{1}{2} \quad \text{for } i \in \{1, 2\}.$$
>
> [...] With this information structure in place we have a Bayesian game of incomplete information in which each player i will choose a signal-dependent action to maximize the probability that a guilty defendant is convicted while an innocent one is acquitted. Because each player has two types, given by the signal he observes, each player will have four pure strategies, $s_i \in \{AA, AC, CA, CC\}$, where strategy xy means that he chooses x when his signal is θ_G and he chooses y when his signal is θ_I.

Note on the book (item 18). Part (a) is true only for $p \leq q$ (the signal precision does not exceed the prior): against a juror who always convicts, a juror who sees θ_I is pivotal and acquits when $p > q$. The book (p. 264) states CC as an equilibrium without the condition. See the solutions PDF, item 18.

(a) What is asked. In the jury game of §12.4 (prior $q > \frac{1}{2}$ of guilt, signals with precision $p > \frac{1}{2}$, payoff 1 for a correct verdict): (a) show that CC for both is a BNE of the two-juror (unanimity) game; (b) compare with a single decider; (c) three jurors, majority rule: when is voting by signal a BNE?

(b) Method. A juror's vote matters only when he is *pivotal*; he must condition his belief about guilt on being pivotal (the swing voter's curse, Tadelis pp. 262–264). Strategy xy: x after signal θ_G, y after θ_I. Posteriors by Bayes' rule: $\Pr(G \mid \theta_G) = \dfrac{qp}{qp + (1-q)(1-p)} > q$ and $\Pr(G \mid \theta_I) = \dfrac{q(1-p)}{q(1-p) + (1-q)p} < q$.

(c) Solution. **(a)** If juror j plays CC, i is pivotal in every state and decides alone. He convicts iff $\Pr(G \mid \theta_i) \geq \frac{1}{2}$. After θ_G this holds; after θ_I it holds iff $q(1-p) \geq (1-q)p$, i.e. iff

$$p \leq q.$$

So (CC, CC) is a BNE *if and only if* $p \leq q$; for $p \leq q$ even the innocent-looking signal is too weak to overturn the prior q, so everyone convicts. *This is a point where the book's statement ("both choosing CC is a BNE", p. 264) needs the condition $p \leq q$:* for $p > q$ a juror with θ_I facing a sure convictor is pivotal and strictly prefers A. For $p > q$ the pure BNE are (AA, AA) (nobody can change an acquittal) and the asymmetric (CA, CC), (CC, CA) (one juror always convicts and the other votes by his signal, being pivotal in every state); voting by signal for both, (CA, CA), is not a BNE for any p, because a juror with θ_I who is pivotal knows the other saw θ_G and, with $\Pr(G \mid \theta_I, \theta_G) = \dfrac{q(1-p)p}{q(1-p)p + (1-q)p(1-p)} = q > \frac{1}{2}$, prefers C.

(b) Payoff = probability of a correct verdict. A single decider who votes by signal gets p (the verdict equals his signal, correct with probability p in each state); always convicting gives q; so he does best with $\max\{p, q\}$: by signal iff $p > q$. In the two-juror game: (CC, CC) gives q (when $p \leq q$), (AA, AA) gives $1 - q < q$ and (CA, CC) gives p (when $p > q$). So a single decider is never worse than any equilibrium of the two-juror game ($\max\{p, q\} \geq q$, p, $1 - q$) and is strictly better when $p > q$ than every *symmetric* equilibrium (the only one is (AA, AA), paying $1 - q$); if one reads the book's (CC, CC) as the equilibrium, the single decider's p beats its q when $p > q$. The reason: under unanimity a juror is pivotal only when the other convicts, so the two jurors cannot both use their information (swing voter's curse), while one decider faces no such problem. The only two-juror equilibria that use information, (CA, CC) and (CC, CA), simply make one juror the single decider.

(c) Three jurors, conviction iff at least two votes C. Suppose the others vote by signal (CA). Juror i is pivotal iff the others split, one θ_G and one θ_I (one votes C, one A): probability $2p(1-p)$ in both states of the world, so the split carries no information about G. Then

$$\Pr(G \mid \theta_i = \theta_G, \text{split}) = \frac{qp}{qp + (1-q)(1-p)} > q > \tfrac{1}{2}, \qquad \Pr(G \mid \theta_i = \theta_I, \text{split}) = \frac{q(1-p)}{q(1-p) + (1-q)p}.$$

After θ_G convicting is optimal; after θ_I acquitting is optimal iff the second posterior is at most $\frac{1}{2}$, i.e. $p \geq q$. So (CA, CA, CA) is a BNE iff

$$p \;\geq\; q,$$

the same condition as for a single decider: when the others' split cancels out, a pivotal juror reads only his own signal.

(d) Interpretation. Unanimity makes the pivotal event "the other juror voted C", which erases the juror's own contrary signal; majority of three makes it "the others disagree", which is uninformative. This is why Feddersen–Pesendorfer find that more than two jurors can restore informative voting.

(e) Verification. `check_E2.py` enumerates all pure profiles of the two-juror game for $(q,p) = (\frac{3}{5},\frac{4}{5})$, $(\frac{4}{5},\frac{3}{5})$, $(\frac{3}{5},\frac{3}{5})$: $p>q$: $\{AA,AA;\ CA,CC;\ CC,CA\}$; $p<q$: $\{AA,AA;\ CC,CC\}$; $p=q$: all four. It asserts that (CC,CC) is a BNE iff $p \le q$, that (CA,CA) never is, the single-decider payoffs p and q and, on an 81-point grid of (q,p), that (CA,CA,CA) is a BNE of the three-juror majority game iff $p \ge q$. Tadelis gives no answer.

Answer. (a) (CC,CC) is a BNE iff $p \le q$ (for $p>q$ it fails). (b) A single decider gets $\max\{p,q\}$: at least as much as any equilibrium of the two-juror game, and strictly more than the symmetric ones when $p>q$ (the unanimity rule wastes the jurors' information through the swing voter's curse). (c) Voting by signal is a BNE of the three-juror majority game iff $p \ge q$.

Item 19: MWG Exercise 13.B.1

Advanced · difficulty 4/5

Statement (verbatim from the list, item 19)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 13 Exercises, Exercise 13.B.1^A, p. 473 [PDF p. 247]. Reproduced verbatim. Figures 13.B.1–13.B.3 are in the text, pp. 440–443. **Answer in the source:** yes, in the MWG solutions manual (Figures 13.B.1(a)-(c)).

> Consider three functions of $\hat{\theta}$: $r(\hat{\theta})$, $E[\theta \mid \theta \leq \hat{\theta}]$, and $\hat{\theta}$. Graph these three functions over the domain $[\underline{\theta}, \bar{\theta}]$, assuming that the first two functions are continuous in $\hat{\theta}$ but allowing them to be otherwise quite arbitrary. Identify the competitive equilibria of the adverse selection model of Section 13.B using this diagram. What about the Pareto optimal labor allocation? Now produce a diagram to depict each of the situations in Figures 13.B.1 to 13.B.3.

(a) What is asked. Graph $r(\hat{\theta})$, $E[\theta \mid \theta \leq \hat{\theta}]$ and $\hat{\theta}$ on $[\underline{\theta}, \bar{\theta}]$; read off the competitive equilibria with unobservable types and the Pareto optimal allocation; then draw the situations of Figures 13.B.1–13.B.3.

(b) Method. Model of MWG §13.B (pp. 437–443): a worker of type θ produces θ at a firm and $r(\theta)$ at home, and accepts a wage w iff $r(\theta) \leq w$. With r increasing, the accepting set at wage $w = r(\hat{\theta})$ is $\{\theta \leq \hat{\theta}\}$ and firms break even iff $w = E[\theta \mid \theta \leq \hat{\theta}]$ (Def. 13.B.1). Reading the diagram in the cutoff $\hat{\theta}$ turns the equilibrium condition into a crossing.

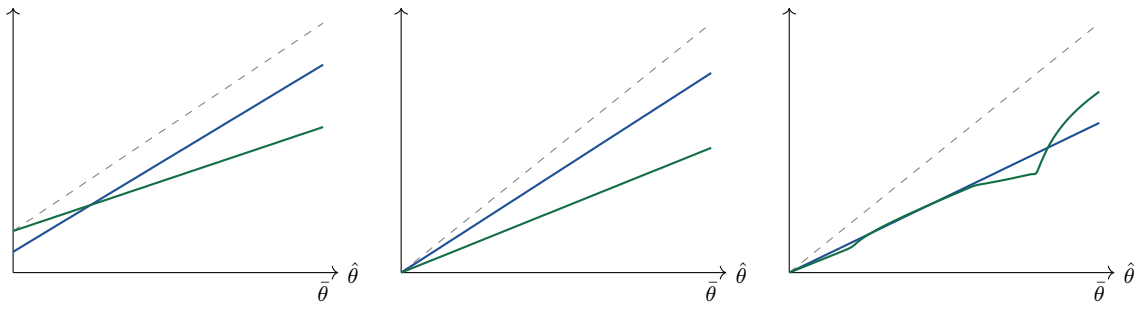
(c) Solution. Competitive equilibria. An interior cutoff $\hat{\theta} \in (\underline{\theta}, \bar{\theta})$ with wage $w = r(\hat{\theta})$ is an equilibrium iff

$$r(\hat{\theta}) = E[\theta \mid \theta \leq \hat{\theta}],$$

a point where the curve $r(\hat{\theta})$ crosses the curve $E[\theta \mid \theta \leq \hat{\theta}]$. Two corners: $\hat{\theta} = \bar{\theta}$ (everybody works, wage $E[\theta]$) is an equilibrium iff $r(\bar{\theta}) \leq E[\theta]$; $\hat{\theta} = \underline{\theta}$ (only the lowest type, a null set, wage $\underline{\theta}$) is an equilibrium iff $r(\underline{\theta}) = \underline{\theta}$ (the curves meet at the left end). The curve $E[\theta \mid \theta \leq \hat{\theta}]$ is increasing (if $f > 0$), starts at $\underline{\theta}$, ends at $E[\theta]$ and lies below the 45° line $\hat{\theta}$.

Pareto optimal allocation. Workers with $r(\theta) \leq \theta$ work at a firm: the set of $\hat{\theta}$ where the curve r lies on or below the 45° line. An equilibrium is efficient only if its cutoff coincides with the boundary of that set.

The three figures (examples checked in `check_E3.py`; blue r, green $E[\theta \mid \theta \leq \hat{\theta}]$, dashed 45°):



- *Fig. 13.B.1* (left): $\theta \sim U[1,2]$, $r(\theta) = 0.9\,\theta \leq \theta$. All types should work, but the unique equilibrium is the crossing $\hat{\theta} = \frac{5}{4}$ (wage $\frac{9}{8}$); here $r(\bar{\theta}) = 1.8 > E[\theta] = 1.5$, so the all-work corner fails: the best workers are driven out.
- *Fig. 13.B.2* (middle): $\theta \sim U[0,2]$, $r(\theta) = 0.8\,\theta$, $\alpha > \frac{1}{2}$: $E[\theta \mid \theta \leq \hat{\theta}] = \hat{\theta}/2 < r(\hat{\theta})$ for every $\hat{\theta} > 0$; the only equilibrium is the corner $\hat{\theta} = \underline{\theta} = 0$, with a null set employed: complete market failure although everybody should work ($r(\theta) < \theta$).

- *Fig. 13.B.3* (right): a density with two bumps ($\theta \in [0,1]$, mass $\frac{1}{20}, \frac{5}{20}, \frac{1}{20}, \frac{13}{20}$ on $[0, \frac{1}{5}], [\frac{1}{5}, \frac{3}{5}], [\frac{3}{5}, \frac{4}{5}], [\frac{4}{5}, 1]$) and $r(\theta) = 0.6\,\theta$: the curves cross three times, at $\hat{\theta} \approx 0.262$, 0.458, 0.834, with wages 0.157, 0.275, 0.501. Wages are Pareto ranked (firms earn 0 everywhere, workers gain with w): the high-wage equilibrium dominates, the low ones are coordination failures.

(d) Interpretation. The three curves summarize the model: efficiency is read against the 45° line (types worth more at the firm than at home), the equilibrium against the expected-productivity curve $E[\theta \mid \theta \leq \hat{\theta}]$, which lies below the 45° line because only the worse types accept low wages: adverse selection.

(e) Verification. `check_E3.py`: sympy solves the crossing for Fig. B.1 ($\hat{\theta} = \frac{5}{4}$, $E = r = \frac{9}{8}$) and shows the $\alpha > \frac{1}{2}$ case of Fig. B.2 has no positive root; it solves the piecewise quadratic equations for Fig. B.3 (three roots, increasing wages), checks that $E[\theta \mid \theta \leq \hat{\theta}]$ is increasing, and writes `fig-mwg13b.tex`. MWG's solutions manual contains the figures; no numbers to compare.

Answer. Equilibria = crossings of $r(\hat{\theta})$ with $E[\theta \mid \theta \leq \hat{\theta}]$ (plus the corners $\hat{\theta} = \bar{\theta}$ if $r(\bar{\theta}) \leq E[\theta]$ and $\hat{\theta} = \underline{\theta}$); the Pareto optimal set of workers is where $r(\hat{\theta}) \leq \hat{\theta}$ (below the 45° line). Fig. B.1: one interior crossing below the efficient set; B.2: only $\hat{\theta} = \underline{\theta}$; B.3: three crossings.

Item 20: MWG Exercise 13.B.2

Advanced · difficulty 4/5

Statement (verbatim from the list, item 20)

Source. Mas-Colell, Whinston & Green (1995), *Microeconomic Theory*, Ch. 13 Exercises, Exercise 13.B.2ˆB, p. 473 [PDF p. 247]. Reproduced verbatim. **Answer in the source:** not found in the solutions manual's OCR text.

> Suppose that $r(\cdot)$ is a continuous and strictly increasing function and that there exists $\hat{\theta} \in (\underline{\theta}, \bar{\theta})$ such that $r(\theta) > \theta$ for $\theta > \hat{\theta}$ and $r(\theta) < \theta$ for $\theta < \hat{\theta}$. Let the density of workers of type θ be $f(\theta)$, with $f(\theta) > 0$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$. Show that a competitive equilibrium with unobservable worker types necessarily involves a Pareto inefficient outcome.

(a) What is asked. With $r(\cdot)$ continuous and strictly increasing, $r(\theta) > \theta$ for $\theta > \theta^\circ$ and $r(\theta) < \theta$ for $\theta < \theta^\circ$ (the exercise's $\hat{\theta}$, renamed to avoid clashing with the cutoff), and $f > 0$: show that any competitive equilibrium with unobservable types is Pareto inefficient.

(b) Method. The Pareto optimal allocation maximizes aggregate surplus (quasilinear preferences): workers with $r(\theta) \le \theta$ work, i.e. $\theta \le \theta^\circ$ (by continuity $r(\theta^\circ) = \theta^\circ$) (MWG pp. 438–439). A competitive equilibrium has a wage w and an accepting set $\{\theta : r(\theta) \le w\} = [\underline{\theta}, \hat{\theta}]$ with $w = E[\theta \mid \theta \le \hat{\theta}]$ (Def. 13.B.1). Show that every equilibrium has $\hat{\theta} < \theta^\circ$ and compare the two sets.

(c) Solution. Because $f > 0$, $E[\theta \mid \theta \le \hat{\theta}] < \hat{\theta}$ for every $\hat{\theta} > \underline{\theta}$. Consider the possible equilibria.

- *Interior cutoff* $\hat{\theta} \in (\underline{\theta}, \bar{\theta})$, $w = r(\hat{\theta})$: the break-even condition gives $r(\hat{\theta}) = E[\theta \mid \theta \le \hat{\theta}] < \hat{\theta}$, hence $r(\hat{\theta}) < \hat{\theta}$, hence $\hat{\theta} < \theta^\circ$ (since $r \ge \theta$ for $\theta \ge \theta^\circ$).
- *Everybody works,* $\hat{\theta} = \bar{\theta}$, $w = E[\theta]$: needs $r(\bar{\theta}) \le E[\theta]$; but $\bar{\theta} > \theta^\circ$ gives $r(\bar{\theta}) > \bar{\theta} > E[\theta]$. Impossible.
- *Only the lowest type (a null set) or nobody works:* the employed set has measure zero.

So in every equilibrium the set of workers employed by firms is $[\underline{\theta}, \hat{\theta}]$ with $\hat{\theta} < \theta^\circ$ (or a null set), while efficiency requires $[\underline{\theta}, \theta^\circ]$. The types in $(\hat{\theta}, \theta^\circ)$, a set of positive measure because $f > 0$, are worth more at a firm ($\theta > r(\theta)$) but stay at home. Aggregate surplus falls short of the maximum by

$$N \int_{\hat{\theta}}^{\theta^\circ} \big(\theta - r(\theta)\big) f(\theta)\, d\theta > 0.$$

With quasilinear utility, transfers can redistribute any surplus gain, so some allocation makes everybody weakly better off and somebody strictly: the equilibrium is not Pareto optimal.

Example. $\theta \sim U[1,2]$, $r(\theta) = 2\theta - \frac{3}{2}$ (so $\theta^\circ = \frac{3}{2}$). The crossing $r(\hat{\theta}) = \frac{1+\hat{\theta}}{2}$ gives $\hat{\theta} = \frac{4}{3} < \frac{3}{2}$, wage $\frac{7}{6}$; the types in $(\frac{4}{3}, \frac{3}{2})$ are lost, and the surplus loss per worker is $\int_{4/3}^{3/2} (\frac{3}{2} - \theta)\, d\theta = \frac{1}{72}$.

(d) Interpretation. The failure is driven by adverse selection: the wage can only equal the *average* productivity of the accepting set, which is below the productivity of the marginal worker, so the marginal worker is never attracted. The situation is the opposite of Fig. 13.B.1–13.B.3 (there all types should work); here only those below θ° should, and still too few do.

(e) Verification. `check_E3.py`: the example (root $\frac{4}{3}$, loss $\frac{1}{72}$, corners fail); and 200 random draws of a positive piecewise-constant density and a linear increasing r crossing θ once: every interior crossing of r and $E[\theta \mid \theta \le \hat{\theta}]$ lies strictly below θ°, and the all-work corner fails. MWG's manual had no OCR text for 13.B.2.

Answer. Every equilibrium has cutoff $\hat{\theta} < \theta^\circ$ (or employs a null set), so the types in $(\hat{\theta}, \theta^\circ)$ with $\theta > r(\theta)$ do not work at a firm; aggregate surplus is strictly below the maximum, hence the equilibrium is Pareto inefficient.

Item 21: Tadelis Exercise 12.7

Advanced · difficulty 4/5

Statement (verbatim from the list, item 21)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 12.7 "Trading Places", p. 268 [PDF p. 287]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Trading Places:** Two players, 1 and 2, each own a house. Each player i values his own house at v_i. The value of player i's house to the other player, i.e., to player $j \neq i$, is $\frac{3}{2}v_i$. Each player i knows the value v_i of his own house to himself, but not the value of the other player's house. The values v_i are drawn independently from the interval $[0, 1]$ with uniform distribution.
>
> a. Suppose players announce simultaneously whether they want to exchange their houses. If both players agree to an exchange, the exchange takes place. Otherwise no exchange takes place. Find a Bayesian Nash equilibrium of this game in pure strategies in which each player i accepts an exchange if and only if the value v_i does not exceed some threshold θ_i.
>
> b. How would your answer to (a) change if player j's valuation of player i's house were $\frac{5}{2}v_i$?
>
> c. Try to explain why any Bayesian Nash equilibrium of the game described in (a) must involve threshold strategies of the type postulated in (a).

(a) What is asked. House swap with private values $v_i \sim U[0,1]$ (house i is worth v_i to its owner and $\frac{3}{2}v_i$ to the other): (a) a threshold BNE; (b) the change if the other's value is $\frac{5}{2}v_i$; (c) why every BNE is a threshold equilibrium.

(b) Method. Player i announces "exchange" iff $v_i \leq \theta_i$. An exchange takes place only if both agree, so i's decision matters only when j agrees; compute i's expected gain from agreeing conditional on that event (Tadelis §12.3, pp. 258–261, same adverse-selection logic).

(c) Solution. **(a)** Suppose j agrees iff $v_j \leq \theta_j$. Then $\Pr(j \text{ agrees}) = \theta_j$ and, when j agrees, $E[v_j] = \theta_j/2$. If i agrees, he gets $\frac{3}{2}v_j$ when j agrees and keeps v_i otherwise; if he refuses he keeps v_i. The gain from agreeing is

$$\theta_j\Big(\tfrac{3}{2}\cdot\frac{\theta_j}{2} - v_i\Big) \geq 0 \iff v_i \leq \tfrac{3}{4}\,\theta_j.$$

This is decreasing in v_i: a threshold rule with $\theta_i = \frac{3}{4}\theta_j$. Simultaneously $\theta_j = \frac{3}{4}\theta_i$, so $\theta_i = \frac{9}{16}\theta_i$ and $\theta_1 = \theta_2 = 0$: the only threshold equilibrium has no one trading (agreeing has probability 0). The reason is adverse selection: a rival who agrees has $E[v_j] = \theta_j/2$, so the house received is worth only $\frac{3}{4}\theta_j < \theta_j$ on average, and each player's cutoff is below the other's.

(b) With $\frac{5}{2}v_j$ the gain is $\theta_j(\frac{5}{2}\cdot\frac{\theta_j}{2} - v_i)$, so $\theta_i = \min\{1, \frac{5}{4}\theta_j\}$. Solutions: $\theta_1 = \theta_2 = 0$, and $\theta_1 = \theta_2 = 1$ (everyone agrees: $\frac{5}{4}\cdot 1 \geq 1$). Interior solutions require $\theta = \frac{25}{16}\theta$, so none with $\theta > 0$. There are two threshold equilibria: no trade, and trade for every pair of types (the swap is then mutually beneficial in expectation: $E[\frac{5}{2}v_j] = \frac{5}{4} > \frac{1}{2} = E[v_i]$).

(c) For a fixed strategy of j with $\Pr(j \text{ agrees}) > 0$, i's gain from agreeing is $\Pr(j \text{ agrees})\big(kE[v_j \mid j \text{ agrees}] - v_i\big)$, a *decreasing linear function* of v_i: i agrees exactly for v_i below a cutoff c. So i's best reply (and hence i's strategy in any BNE) is a threshold rule. The exception is a j who never agrees: then i's choice is irrelevant and any rule is a best reply, but j's own behavior (never agreeing) is the threshold $\theta_j = 0$.

(d) Interpretation. (a) is Akerlof's lemons logic with two-sided private information: gains from trade exist for every pair ($\frac{3}{2}v_j > v_j$) and still no trade survives, because agreeing is informative about the value of the house one receives. (b) Increasing the gain from the swap supports a full-trade equilibrium as well, with no-trade remaining another equilibrium.

(e) Verification. `check_E3.py`: sympy checks $E[v_j \mid v_j \le \theta_j] = \theta_j/2$; scanning a grid of thresholds for the fixed points of $\theta \mapsto \min\{1, k\theta/2\}$ returns $\{(0,0)\}$ for $k = \frac{3}{2}$ and $\{(0,0),(1,1)\}$ for $k = \frac{5}{2}$; checks that $k/2 \cdot k/2 \neq 1$ (no interior solutions); and that the gain function is monotone decreasing in v_i. Tadelis gives no answer.

Answer. (a) The only threshold equilibrium is $\theta_1 = \theta_2 = 0$ (no trade). (b) Thresholds $\theta_i = \min\{1, \frac{5}{4}\theta_j\}$: two equilibria, $(0,0)$ and $(1,1)$ (full trade). (c) The gain from agreeing is decreasing and linear in v_i, so best replies are cutoff rules.

Item 22: Gibbons Problem 3.8

Advanced · difficulty 5/5

Statement (verbatim from the list, item 22)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §3.5 Problems, Section 3.2, Problem 3.8, pp. 171–172 [PDF pp. 177–178]. Reproduced verbatim. Figure 3.2.3 (the double auction's trading region) is in the text, p. 163. **Answer in the source:** no.

> **3.8.** Reinterpret the buyer and seller in the double auction analyzed in Section 3.2.C as a firm that knows a worker's marginal product (m) and a worker who knows his or her outside opportunity (v), respectively, as in Hall and Lazear (1984). In this context, trade means that the worker is employed by the firm, and the price at which the parties trade is the worker's wage, w. If there is trade, then the firm's payoff is $m - w$ and the worker's is w; if there is no trade then the firm's payoff is zero and the worker's is v.
>
> Suppose that m and v are independent draws from a uniform distribution on $[0, 1]$, as in the text. For purposes of comparison, compute the players' expected payoffs in the linear equilibrium of the double auction. Now consider the following two trading games as alternatives to the double auction.
>
> *Game I*: Before the parties learn their private information, they sign a contract specifying that if the worker is employed by the firm then the worker's wage will be w, but also that either side can escape from the employment relationship at no cost. After the parties learn the values of their respective pieces of private information, they simultaneously announce either that they Accept the wage w or that they Reject that wage. If both announce Accept, then trade occurs; otherwise it does not. Given an arbitrary value of w from $[0, 1]$, what is the Bayesian Nash equilibrium of this game? Draw a diagram analogous to Figure 3.2.3 showing the type-pairs that trade. Find the value of w that maximizes the sum of the players' expected payoffs and compute this maximized sum.
>
> *Game II*: Before the parties learn their private information, they sign a contract specifying that the following dynamic game will be used to determine whether the worker joins the firm and if so at what wage. (Strictly speaking, this game belongs in Chapter 4. We will anticipate the spirit of Chapter 4 by arguing that this game can be solved by combining the lessons of this chapter with those of Chapter 2.) After the parties learn the values of their respective pieces of private information, the firm chooses a wage w to offer the worker, which the worker then accepts or rejects. Try to analyze this game using backwards induction, as we did for the analogous complete-information games in Section 2.1.A, as follows. Given w and v, what will the worker do? If the firm anticipates what the worker will do, then given m what will the firm do? What is the sum of the players' expected payoffs?

Referred to in the statement. The double auction of Section 3.2.C, its linear equilibrium and Figure 3.2.3 (Gibbons pp. 159, 161 and 163 [PDF p. 165, 167 and 169]) (reproduced from the book; the figure is described in words):

> We analyze a trading game called a double auction. The seller names an asking price, p_s, and the buyer simultaneously names an offer price, p_b. If $p_b \geq p_s$, then trade occurs at price $p = (p_b + p_s)/2$; if $p_b < p_s$, then no trade occurs.
>
> The buyer's valuation for the seller's good is v_b, the seller's is v_s. These valuations are private information and are drawn from independent uniform distributions on $[0, 1]$. If the buyer gets the good for price p, then the buyer's utility is $v_b - p$; if there is no trade, then the buyer's utility is zero. If the seller sells the good for price p, then the seller's utility is $p - v_s$; if there is no trade, then the seller's utility is zero. (Each of these utility functions measures the change in the party's utility. If there is no trade, then there is no change in utility. It would make no difference to define, say, the seller's utility to be p if there is trade at price p and v_s if there is no trade.)
>
> [...] Therefore, the linear equilibrium strategies are
>
> $$p_b(v_b) = \frac{2}{3}v_b + \frac{1}{12} \tag{3.2.7}$$

and

$$p_s(v_s) = \frac{2}{3}v_s + \frac{1}{4}, \tag{3.2.8}$$

as shown in Figure 3.2.2.

Recall that trade occurs in the double auction if and only if $p_b \geq p_s$. Manipulating (3.2.7) and (3.2.8) shows that trade occurs in the linear equilibrium if and only if $v_b \geq v_s + (1/4)$, as shown in Figure 3.2.3.

Figure 3.2.3 (described): the unit square with v_s on the horizontal axis and v_b on the vertical axis; it draws the line $v_b = v_s$ and the line $v_b = v_s + (1/4)$, and labels the region above $v_b = v_s + (1/4)$ "TRADE".

(a) What is asked. Firm (marginal product m) and worker (outside opportunity v), m, v independent $U[0,1]$: expected payoffs in the linear equilibrium of the double auction; then Game I (fixed wage w, each side accepts or rejects) and Game II (the firm offers a wage): equilibrium, trading region, and the sum of expected payoffs.

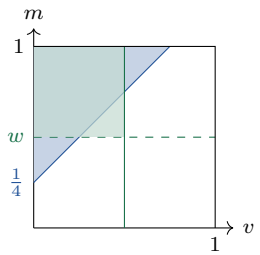
(b) Method. Gibbons §3.2.C (pp. 158–163). Trade gives the firm $m - w$ and the worker w; no trade gives 0 and v. Because v is the worker's payoff without trade, the *sum of expected payoffs* is $E[v] + E[(m-v)\mathbf{1}_{\text{trade}}] = \frac{1}{2} + (\text{gains from trade})$.

(c) Solution. Double auction. Worker (seller) names p_s, firm (buyer) names p_b; trade at $(p_b + p_s)/2$ iff $p_b \geq p_s$. With linear strategies $p_b = a_b + c\,m$, $p_s = a_s + c\,v$ the firm's FOC gives $p_b = \frac{2}{3}m + \frac{1}{3}a_s$ and the worker's $p_s = \frac{2}{3}v + \frac{1}{3}(a_b + c)$; so $c = \frac{2}{3}$, $a_b = \frac{1}{12}$, $a_s = \frac{1}{4}$:

$$p_b(m) = \tfrac{2}{3}m + \tfrac{1}{12}, \qquad p_s(v) = \tfrac{2}{3}v + \tfrac{1}{4}, \qquad \text{trade} \iff p_b \geq p_s \iff m \geq v + \tfrac{1}{4}.$$

Trade probability $\frac{9}{32}$; gains from trade $E[(m-v)\mathbf{1}] = \frac{9}{64}$; price $w = \frac{m+v}{3} + \frac{1}{6}$. Expected payoffs: firm $E[(m-w)\mathbf{1}] = \frac{9}{128}$, worker $\frac{1}{2} + E[(w-v)\mathbf{1}] = \frac{73}{128}$; sum $\frac{82}{128} = \frac{41}{64}$ (gains $\frac{9}{64} \approx 0.141$).

Game I (fixed w). After learning their information the firm accepts iff $m \geq w$ and the worker iff $w \geq v$ (accepting weakly dominates; the trade-never equilibrium in which both always reject is a Nash equilibrium too, but uses weakly dominated actions). Trade occurs iff $m \geq w \geq v$: the rectangle $[0,w] \times [w,1]$ in (v,m) space, versus the double auction's triangle $m \geq v + \frac{1}{4}$.



Trading regions in (v,m) space: blue triangle = double auction; green rectangle = Game I for $w = \frac{1}{2}$ (the two overlap).

Gains from trade at wage w: $P(m \geq w)P(v \leq w)\big[E(m \mid m \geq w) - E(v \mid v \leq w)\big] = (1-w)\,w \cdot \frac{1}{2}$, maximized at $w = \frac{1}{2}$ with value $\frac{1}{8}$, so the maximized sum of expected payoffs is $\frac{1}{2} + \frac{1}{8} = \frac{5}{8}$. At $w = \frac{1}{2}$ the firm earns $\frac{1}{16}$ and the worker $\frac{1}{2} + \frac{1}{16}$. This is below the double auction's $\frac{9}{64}$ in gains (and $\frac{41}{64}$ in sum).

Game II (the firm offers). The worker accepts w iff $w \geq v$, with probability w. The firm with marginal product m maximizes $(m-w)\,w$: $w = m/2$ (the payoff $m^2/4$). Trade occurs iff $v \leq m/2$. Gains from trade $\int_0^1 \int_0^{m/2} (m-v)\,dv\,dm = \int_0^1 \frac{3}{8}m^2 dm = \frac{1}{8}$; firm $\frac{1}{12}$, worker $\frac{1}{2} + \frac{1}{24}$; sum $\frac{5}{8}$.

(d) Interpretation. The double auction is best (gains $\frac{9}{64} > \frac{1}{8}$), though not efficient (first best $E[(m-v)^+] = \frac{1}{6}$): its regions of trade are not rectangular, so it adapts the trade rule to both types. Game II (informed principal) yields the same $\frac{1}{8}$ as the best fixed wage.

(e) Verification. `check_E3.py` (sympy): the linear equilibrium coefficients and best-reply FOC; the trade condition $m \geq v + \frac{1}{4}$; the integrals $\frac{9}{32}$, $\frac{9}{64}$, $\frac{9}{128}$, $\frac{73}{128}$, $\frac{41}{64}$; Game I gains $w(1-w)/2$, firm $w(1-w)^2/2$, worker $\frac{1}{2} + (1-w)w^2/2$, the maximum at $w = \frac{1}{2}$; Game II offer $m/2$, gains $\frac{1}{8}$, payoffs $\frac{1}{12}$, $\frac{1}{2} + \frac{1}{24}$. Gibbons gives no answer.

Answer. Double auction: firm $\frac{9}{128}$, worker $\frac{73}{128}$, sum $\frac{41}{64}$ (gains $\frac{9}{64}$). Game I: trade iff $m \geq w \geq v$; best $w = \frac{1}{2}$, sum $\frac{5}{8}$ (gains $\frac{1}{8}$). Game II: the firm offers $w = m/2$; trade iff $v \leq m/2$; sum $\frac{5}{8}$ (gains $\frac{1}{8}$).

Item 23: Tadelis Exercise 13.5

Advanced · difficulty 5/5

Statement (verbatim from the list, item 23)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 13.5 "Mixed Strategies", p. 286 [PDF p. 305]. Reproduced verbatim. **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Mixed Strategies:** Consider a seller who offers a single private value good using a first-price sealed-bid auction. There are two potential buyers, each with a valuation that can take on one of two values, $\theta_i \in \{4, 8\}$, each value occurring with an equal probability of $\frac{1}{2}$. The players' values are independently drawn. Call the type with $\theta_i = 4$ the "low" type and $\theta_i = 8$ the "high" type.
>
> a. Show that in a Bayesian Nash equilibrium the low must be using a pure strategy bid $b_L = 4$. (Hint: First show that in a Bayesian Nash equilibrium the low type cannot be using a pure strategy bid $b_L < 4$. Next show that in a Bayesian Nash equilibrium the low type cannot be using a mixed strategy with two different bids.)
>
> b. Show that given the fact described in (a), in any Bayesian Nash equilibrium the high type will never choose a bid $b_H > 6$.
>
> c. Show that the low type choosing $b_L = 4$ and the high type choosing b_H uniformly from the interval $[4, 6]$ together constitute a Bayesian Nash equilibrium.
>
> d. Calculate the expected revenue to the seller given the Bayesian Nash equilibrium in (c). How does it compare to the expected revenue he would obtain from an English auction?

Note on the book (item 23). In part (c) the uniform distribution on $[4, 6]$ is not an equilibrium strategy of

the high type (the payoff $(8 - b)(b - 2)/4$ is not constant on $[4, 6]$); the indifference condition gives the cdf $(b - 4)/(8 - b)$ on $[4, 6]$, whose revenue equals the English auction's 5. See the solutions PDF, item 23.

(a) What is asked. First-price auction, two bidders, values $\theta_i \in \{4, 8\}$ equiprobable: (a) the low type bids 4; (b) the high type never bids above 6; (c) check the proposed equilibrium; (d) revenue against the English auction.

(b) Method. Symmetric BNE, ties broken by a fair coin. Winning probabilities are weakly increasing in the bid; a mixed strategy needs equal payoffs on its support (Tadelis §6.3, §13.1.3). Let $W(b)$ be the probability that a bid b wins.

(c) Solution. **(a)** A low type never bids above 4 (winning there pays more than 4; bid 4 already guarantees at least 0), and the bid 4 earns exactly 0. *No atom below* 4. Suppose a bid $\beta < 4$ is used with positive probability. A rival low type then also bids β with positive probability, so β ties with positive probability. Raising the bid to $\beta + \varepsilon$ turns each such tie into a sure win, a discrete gain in W, at cost at most ε; so for small ε the deviation pays, contradicting optimality. This rules out a pure $b_L < 4$ and any mixture over two different bids below 4. *No atomless mixing below* 4. If the low type's support were an atomless set below 4, with infimum $\underline{b} < 4$ and supremum $\bar{b} \leq 4$, the lowest bids win with probability close to 0 (they can beat only rival low types bidding even lower), so their payoff tends to 0, whereas $\bar{b} < 4$ beats every rival low type and earns at least $\frac{1}{2}(4 - \bar{b}) > 0$; the two cannot be equal. If $\bar{b} = 4$ the same argument applies at $\underline{b}$. Hence the low type's bid is the pure bid $b_L = 4$.

(b) Given $b_L = 4$, a high type bidding $4 + \varepsilon$ beats every rival low type (probability $\frac{1}{2}$), so his payoff is at least $\frac{1}{2}(4 - \varepsilon) \to 2$. A bid $b > 6$ earns at most $8 - b < 2$ (the price is at most b and winning probability is at most 1), which is strictly less than $\frac{1}{2}(4 - \varepsilon)$ for ε small. So no bid above 6 is optimal.

(c) *The uniform distribution on* $[4,6]$ *is not an equilibrium strategy of the high type; the equilibrium has cdf* $G(b) = \dfrac{b-4}{8-b}$ *on* $[4,6]$. A high type bidding $b \in (4,6]$ wins against all low types (probability $\frac{1}{2}$) and against a high type bidding below b (probability $\frac{1}{2}G(b)$):

$$U_H(b) = (8-b)\Big(\tfrac{1}{2} + \tfrac{1}{2}G(b)\Big).$$

Mixing requires U_H constant on the support. As $b \downarrow 4$, $U_H \to 4 \cdot \frac{1}{2} = 2$ (a bid exactly 4 ties with low types and earns only 1), so $U_H \equiv 2$ gives $1 + G(b) = \dfrac{4}{8-b}$, i.e. $G(b) = \dfrac{b-4}{8-b}$, with $G(4) = 0$, $G(6) = 1$ and density $\dfrac{4}{(8-b)^2}$. With the *uniform* $G(b) = (b-4)/2$ the payoff is $U_H(b) = \frac{(8-b)(b-2)}{4}$, equal to 2 at $b = 4^+$ and $b = 6$ but $\frac{9}{4}$ at $b = 5$: the high type would deviate to a pure bid near 5, so uniform mixing is not a BNE (the book's wording in (c) is imprecise on this point). With the correct G: the high type is indifferent over $[4,6]$ (payoff 2), strictly worse above 6 ($8 - b < 2$) and strictly worse below 4 (payoff 0); the low type earns 0 at $b_L = 4$ and $(4-b)W(b) < 0$ at any $b > 4$. So $b_L = 4$ and $b_H \sim G$ form a BNE.

(d) Revenue. Both low ($\frac{1}{4}$): the price is 4. One high ($\frac{1}{2}$): the high type wins and pays $b_H \sim G$, $E[b_H] = \int_4^6 \frac{4b}{(8-b)^2} db$. Both high ($\frac{1}{4}$): the maximum of two draws, cdf G^2. Evaluating,

$$\text{Rev} = \tfrac{1}{4} \cdot 4 + \tfrac{1}{2}E[b_H] + \tfrac{1}{4}E[\max] = 5.$$

The English auction gives the second-highest value: 8 if both are high ($\frac{1}{4}$), else 4: $\frac{1}{4} \cdot 8 + \frac{3}{4} \cdot 4 = 5$. The two are equal: revenue equivalence. (With the uniform on $[4,6]$ one would get $\frac{29}{6} < 5$, another sign that it is not the equilibrium.)

(d) Interpretation. With only two values, the high type must randomize to avoid being undercut by a pure bid; the support starts at the low type's value because bidding below it never wins against a low type's bid of 4, and the shading is exactly enough to make revenue equal to the English auction's.

(e) Verification. `check_E3.py`: sympy verifies $G(4) = 0$, $G(6) = 1$, $U_H \equiv 2$ on $(4,6]$; shows $U_H^{\text{unif}}(5) = \frac{9}{4} > 2$; integrates $E[b_H]$ and $E[\max]$ to obtain revenue exactly 5 = English revenue; and the low type's strictly negative payoff from bids above 4. Tadelis gives no answer.

Answer. (a) $b_L = 4$. (b) No high bid above 6. (c) The high type mixes on $[4,6]$ with cdf $G(b) = \frac{b-4}{8-b}$ (the uniform fails: $U_H(5) = \frac{9}{4} > 2$). (d) Revenue 5, equal to the English auction's $\frac{1}{4} \cdot 8 + \frac{3}{4} \cdot 4 = 5$.

Item 24: Tadelis Exercise 13.10

Advanced · difficulty 4/5

Statement (verbatim from the list, item 24)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 13.10 "Common Values Linear Bidding", p. 287 [PDF p. 306]. Reproduced verbatim. "The previous exercise" is item 16 (Tadelis 13.9). **Answer in the source:** no (none in the book; Princeton UP's website has about half, not checked).

> **Common Values Linear Bidding:** Two players participate in a first-price sealed-bid auction in a common-values setting. The value of the good for each player is given by $\theta_1 + \theta_2$, where θ_i is a signal observed only by player i and is uniformly distributed over the interval $[0, 1]$. Hence if player i observes his signal θ_i then his expected valuation is equal to $0.5 + \theta_i$ (but unlike the previous exercise, this is a common-values setting).
>
> a. Show that there is a symmetric Bayesian Nash equilibrium in which each bidder i's bidding strategy is of the form $s_i^*(\theta_i) = a + b\theta_i$.
>
> b. What is the expected payoff of a player if his type is θ_i?
>
> c. Show that for each value of θ_i the equilibrium bid of player i is less than what you found in (a) of the previous exercise.

Referred to in the statement. The previous exercise, Exercise 13.9 "Private Value Linear Bidding" (Tadelis p. 287 [PDF p. 306]) (reproduced from the book):

> **Private Value Linear Bidding:** Two players participate in a first-price sealed-bid auction in an IPV setting. The value of the good for each player is given by $0.5 + \theta_i$, where θ_i is observed only by player i and is uniformly distributed over the interval $[0, 1]$.
>
> a. Show that there is a symmetric Bayesian Nash equilibrium in which each bidder i's bidding strategy is of the form $s_i^*(\theta_i) = a + b\theta_i$.
>
> b. What is the expected payoff of a player if his type is θ_i?

Optional extras.

- Tadelis 12.4 "Grade Gambles" (p. 267): a finite Bayesian game where more information hurts; pair it with Muñoz-Garcia Ch. 7 Ex. 4 "More Information Might Hurt" (p. 331).

- Tadelis 12.10 "Purification" (p. 269): uniqueness of the perturbed Matching Pennies BNE, after item 12; Gibbons 3.5 (p. 170) asks for the same construction.

- Gibbons 3.2 (p. 169) and Muñoz-Garcia Ch. 7 Ex. 5 (p. 336): Cournot and differentiated duopoly with private demand information, after item 9.

- Muñoz-Garcia Ch. 7 Ex. 6 (p. 339, public good) and Ex. 7 (p. 343, starting a fight): more threshold BNE, with answers.

- Muñoz-Garcia Ch. 8 Ex. 2–3 (pp. 352, 355): first-price auction with N bidders and general distributions, after item 15; Gibbons 3.7 (pp. 170–171) is the two-bidder general-density case.

- Tadelis 13.6 (pp. 286–287) and Muñoz-Garcia Ch. 8 Ex. 7 (p. 371): the third-price auction and revenue equivalence.

- Tadelis 13.8 (p. 287): the oil-field second-price auction with common values, after item 24.

- Tadelis 13.4 (p. 286) and 13.7 (p. 287); Muñoz-Garcia Ch. 8 Ex. 5–6 (pp. 364–365, all-pay) and Ex. 8 (p. 371, risk-averse bidders): beyond the lecture's core.

(a) What is asked. Two bidders, first-price auction, value $\theta_1 + \theta_2$ with private signals $\theta_i \sim U[0,1]$: (a) a symmetric BNE $s(\theta_i) = a + b\theta_i$; (b) the expected payoff of type θ_i; (c) compare the bid with item 16.

(b) Method. Common values: winning is informative about the rival's signal (the winner's curse; Tadelis §13.2, pp. 282–284). Let the rival bid $a + b\theta_j$ with $b > 0$; a bid x wins iff $\theta_j < t \equiv (x-a)/b$, and then $E[\theta_j \mid \theta_j < t] = t/2$.

(c) Solution. **(a)** Given signal θ_i and a bid $x = a + bt$,

$$EU_i = \underbrace{t}_{\Pr(\text{win})} \Big(\theta_i + \frac{t}{2} - a - bt\Big).$$

FOC in t: $\theta_i - a + t - 2bt = 0$, so $t = \dfrac{\theta_i - a}{2b-1}$ (second derivative $1 - 2b < 0$ needs $b > \frac{1}{2}$). In a symmetric equilibrium the bid of type θ_i beats exactly the rival types below θ_i, i.e. $t = \theta_i$ for all θ_i:

$$(2b-1)\theta_i = \theta_i - a \;\; \forall \theta_i \;\Rightarrow\; 2b - 1 = 1,\; a = 0 \;\Rightarrow\; b = 1,\; a = 0.$$

So $s^*(\theta_i) = \theta_i$: bid one's own signal. Conditional on winning the item is worth $\theta_i + \theta_i/2$ (the rival's signal is lower), and the bid shades that by $\theta_i/2$.

(b) With $t = \theta_i$: $EU_i^* = \theta_i\big(\theta_i + \frac{\theta_i}{2} - \theta_i\big) = \dfrac{\theta_i^2}{2}$.

(c) Item 16 (private values $0.5 + \theta_i$) has $s(\theta_i) = \frac{1}{2} + \frac{1}{2}\theta_i$. Then

$$\tfrac{1}{2} + \tfrac{1}{2}\theta_i - \theta_i = \tfrac{1}{2}(1 - \theta_i) > 0 \quad \text{for } \theta_i < 1,$$

so the common-value bid is strictly lower for every $\theta_i < 1$ (they coincide at $\theta_i = 1$, a single type). Winner's curse: conditional on winning, the rival's signal is lower than θ_i, so the item is worth less than $0.5 + \theta_i$; bidders shade more.

(d) Interpretation. Both exercises give the same expected payoff $\theta_i^2/2$, but with common values the bid is lower because winning is bad news about the rival's signal.

(e) Verification. `check_E3.py`: sympy solves the FOC, finds $(a, b) = (0, 1)$ by matching, checks strict concavity (-1 at $b = 1$) and the payoff $\theta^2/2$; scipy maximizes EU_i against $s(\theta) = \theta$ at $\theta = 0.1, 0.4, 0.8$ and returns bids θ and value $\theta^2/2$; compares with $\frac{1}{2} + \frac{1}{2}\theta$. Tadelis gives no answer.

Answer. (a) $s^*(\theta_i) = \theta_i$ ($a = 0$, $b = 1$). (b) $EU_i^*(\theta_i) = \theta_i^2/2$. (c) The bid is lower than item 16's $\frac{1}{2} + \frac{1}{2}\theta_i$ by $\frac{1}{2}(1 - \theta_i)$, strictly for $\theta_i < 1$ (winner's curse).